



2025 Regional Transmission Plan

Executive Summary

The 2025 Regional Transmission Plan (RTP) is the result of a coordinated planning process performed by ERCOT Grid Planning with extensive review and input by Transmission Planners (TPs) and Transmission Owners (TOs) registered with North American Electric Reliability Corporation (NERC), and other stakeholders. The 2025 RTP addresses ERCOT System transmission needs for years 2027 through 2031, based on the assessment of various factors, including reliability and economic analyses. This report documents the results of the assessment, in part, to comply with the requirements of NERC Reliability Standards, ERCOT Protocols, and the ERCOT Planning Guide.

The reliability analysis was performed over a six-year planning horizon; years two through five representing the near-term horizon and year six representing the long-term horizon. The 2025 RTP assessed ERCOT's steady-state transmission needs under summer peak and off-peak conditions. In addition to the seasonal variations, the 2025 RTP also included sensitivity analyses to address uncertainty involved in the transmission planning process. The reliability analysis in the 2025 RTP included:

- Steady-state contingency analysis to identify criteria violations based on NERC Reliability Standards and ERCOT planning criteria
- Short circuit analysis to identify over-dutied circuit breakers in the near-term planning horizon
- Cascading analysis to identify potential system cascading conditions

Following the reliability analysis, ERCOT, in collaboration with TPs, developed Corrective Action Plans (CAPs) to address the reliability criteria violations identified. These plans included, but were not limited to, upgrades or the addition of new transmission facilities and new Constraint Management Plans (CMPs).

The ERCOT System is experiencing rapid changes, including trends of notable growth in demand and penetration of intermittent Generation Resources. This trend of rising demand is expected to continue, driven by factors such as the further electrification of oil and gas processes in the Permian Basin and continued interest in connecting large loads to the ERCOT System. Sufficient transmission capacity plays a key role in ensuring system reliability while meeting the rapid demand growth. While transmission still requires three to five years to be energized, demand is escalating as new large loads are added to the ERCOT System faster (six to 12 months) and in greater sizes than historical interconnections.

As the new load interconnection timeline outpaces the development of transmission projects, the Texas Legislature recognized the need to update rules in the ERCOT Region. Previously, these rules did not allow ERCOT to include loads in planning studies without signed interconnection agreements. In December 2022, the Public Utility Commission of Texas (PUCT) amended 16 Texas Administrative Code (TAC) § 25.101, Certification Criteria, to implement Senate Bill 1281 (87th Legislative Session). The amendment included the requirement for any review conducted by ERCOT to incorporate

historical load, forecasted load growth, and additional load currently seeking interconnection. In June 2023, House Bill (HB) 5066 (88th Legislative Session) was enacted, introducing a new requirement to consider loads for which the electric utility has yet to sign an interconnection agreement, as determined by the electric utility responsible for serving the load.

These new requirements were first implemented in the 2024 RTP and led to a significant increase in large loads included in that assessment (i.e., data centers, crypto mining, hydrogen and hydrogen-related manufacturing, and electrification). In the 2024 RTP, the forecasted summer peak demand for 2030 exceeded 150 gigawatts (GW), with close to 40 GW attributed to large loads seeking interconnections under the new requirements.

The large loads submitted by Transmission Service Providers (TSPs) for consideration under these requirements in the 2025 RTP were significantly greater compared to the 2024 RTP. The forecasted summer peak demand for 2031 would have exceeded 218 GW if all TSP-submitted large loads were included. After performing a historical analysis on large load realization, ERCOT developed an ERCOT Transmission Planning Adjusted Load Forecast (ETPALF), which adjusted the 2031 forecast from 218 GW to 159 GW. The ETPALF preserved large loads substantiated with an interconnection agreement at 100%; large loads substantiated with officer attestations (“Officer Letters”) were reduced to 55.4%; and, data center officer letter loads (OLLs) were further reduced to 49.8% of the initial OLL reduction, resulting in a final amount that was 27.6% of the full data center value.

On May 1, 2025, ERCOT filed a Good Cause Exception (GCE)¹ to the PUCT requesting use of the ETPALF in the 2025 RTP. The PUCT approved the GCE on June 5, 2025. ERCOT received final load substantiations from certificated Load Serving Entities at the end of June, and it was not until August 5, 2025, that the start cases for the 2025 RTP were finalized and posted. The delay in the finalization and posting of the start cases resulted in insufficient time for ERCOT to complete all required components of the 2025 RTP by the end of the 2025 as required by ERCOT Planning Guide Section 3.1.1.2(2). Accordingly, on November 12, 2025, ERCOT filed a second GCE² requesting approval to not perform the Maintenance Outage analysis³ for 2025 RTP as required by ERCOT Planning Guide Section 4.1.1.8. The PUCT approved the GCE request on December 12, 2025.

While the new requirements identified above allow for the consideration of more load in the planning processes, the policy governing the inclusion of planned Generation Resources in the planning models remains unchanged. Currently, planned Generation Resources can only be added to the planning models when they meet the requirements established under ERCOT Planning Guide Section 6.9, Addition of Proposed Generation to the Planning Models. Under Planning Guide Section 6.9 criteria, the total available generation was approximately 130 GW in the 2031 summer peak case based on the 2025 RTP renewable and battery dispatch assumptions. This indicated a supply deficit of approximately 30 GW to meet the demand. Additional steps outside of the current planning practices

¹ https://interchange.puc.texas.gov/Documents/55999_121_1495046.PDF

² https://interchange.puc.texas.gov/Documents/55999_174_1557350.PDF

³ Maintenance Outage analysis is an ERCOT regional requirement above the NERC TPL requirements

were needed to create planning models capable of accommodating the unprecedented load growth. To achieve a balance between supply and demand, ERCOT incorporated generators that had not yet met the Planning Guide Section 6.9 requirements into the planning models and presented the assumptions⁴ at the July 2025 Regional Planning Group (RPG) stakeholder meeting. ERCOT is currently working with stakeholders on Planning Guide Revision Request (PGRR) 127 to adjust the criteria for adding Generation Resources to the planning models and is expected to complete the stakeholder process in early 2026.

The recent unprecedented load growth first sparked discussions in the 2024 RTP about introducing an Extra High Voltage (EHV) 765-kilovolt (kV) backbone infrastructure to the ERCOT Region. This EHV backbone would function as a superhighway, enabling electrons to flow more efficiently through long-distance transmission from resource-rich regions to load centers. The benefits of higher voltage transmission include:

- Increased transfer capability to load centers
- Flexibility in Generation Resource siting
- Increased outage coordination capacity
- Reduced impact to Texas consumers due to less overall Right-of-Way requirements⁵
- Reduced line losses
- Potential retirement of series compensation devices
- Potential exit strategy for some current Generic Transmission Constraints

In July 2024, ERCOT filed the Permian Basin Reliability Plan⁶ with the PUCT, which identified various 345-kV, 500-kV, and 765-kV import path options to address the reliability needs in the Permian Basin area. The PUCT approved the Permian Basin Reliability Plan in October 2024, and on May 1, 2025, approved the 765-kV import option, introducing a higher transmission voltage level in ERCOT for the first time since the 1960s. In December 2025, ERCOT's Board of Directors approved the 765-kV Eastern Backbone and Western Loop projects, thereby approving the full 765-kV Strategic Transmission Expansion Plan (STEP)⁷.

⁴ https://www.ercot.com/files/docs/2025/07/29/2025_RTP_Assumptions_July_2025_RPG.pdf

⁵ A single 765-kV circuit requires approximately 200 feet of Right-of-Way but carries up to three times the power of a 345-kV double circuit. A single 345-kV double circuit requires approximately 150 feet of Right-of-Way. When comparing overall Right-of-Way needs, the 765-kV option requires less than half (200 feet) what the equivalent 345-kV option requires (150 feet x 3 double circuits = 450 total feet for equivalent 345-kV transmission capacity).

⁶

<https://interchange.puc.texas.gov/search/filings/?UtilityType=A&ControlNumber=55718&ItemMatch=Equal&DocumentType=ALL&SortOrder=Ascending>

⁷ <https://www.ercot.com/files/docs/2025/01/27/2024-regional-transmission-plan-rtp-345-kv-plan-and-texas-765-kv-strategic-transmission-expans.pdf>

The 765-kV STEP was included in the 2025 RTP and showed continued reliability benefits seen in the 2024 RTP. The 2025 RTP evaluated the potential addition of 765-kV equipment beyond what is already included in the STEP to address new load growth, and additional 765-kV equipment was ultimately added to address a growing import need in Central Texas.

Unprecedented load growth, insufficient generation to meet this demand, and the faster pace of load interconnections compared to transmission and generation development moved the ERCOT Region into a new era of planning. These challenges are further compounded by ongoing changes in the resource mix. The locations of both load and generation are crucial in driving flow patterns and determining transmission needs. While the generation interconnection queue offers some insight into future generation development, the location of future loads is less predictable, especially given the much shorter lead times for load interconnections. This adds complexity to transmission planning, as flow patterns and stressed import paths may change between planning cycles. Planning practices and policies must continue to be reviewed to address these new challenges. ERCOT will continue to work with stakeholders to identify the necessary changes in the planning processes to ensure system reliability.

In addition, the rapid growth of wind, solar, and energy storage resources, coupled with increased retirements of coal and gas generators, led to greater reliance on intermittent, renewable generation to meet the higher system demand. This resulted in more long-distance power transfers as renewable generation is typically located far from load centers. Similar to the 2023 and 2024 RTPs, the 2025 RTP assessment results highlighted the ongoing need for major import pathway enhancements to the Central Texas corridor. However, with the continued addition of significant amounts of large load in and around the Corpus Christi area in the 2025 RTP (as first seen in the 2024 RTP), imports to the Central Texas area primarily come from the Coast Weather Zone, specifically Southwest of Houston, instead of from South Texas as indicated in the 2023 RTP assessment results.

Overall, 192 reliability projects were identified to address all reliability violations in the 2025 RTP which included the 765-kV STEP in its base case, compared to 145 projects in the 2024 RTP with the 765-kV STEP in its base case, and 173 projects in the 2023 RTP. All RTPs from 2023 and earlier did not include 765-kV STEP in their base cases. This trend emphasizes the transmission challenges associated with the rapidly changing grid.

The majority of planned improvements identified in the 2025 RTP are 138-kV and 345-kV system upgrades. The projects identified as 345-kV upgrades consist of new substations, transmission line additions, upgrades and rebuilds, new 345/138-kV transformers, existing 345/138-kV transformer upgrades, and the addition of capacitor banks and reactors.

The 2025 RTP identified the following noteworthy reliability projects, some of which were also identified in the 2024 RTP:

- Horseshoe Spring Switch 345/138-kV substation expansion with two 345/138-kV transformers in Pecos County
- Scurry County South Switch to Galvani 345-kV double-circuit line upgrade in Scurry County

- HODL Ranch to Clear Crossing 345-kV line upgrade in Haskell County
- Carmichael Bend Switch to Benbrook Switch 345-kV line upgrades in Hood and Tarrant Counties
- Royse area 345-kV line upgrades in Collin, Rockwall, Fannin, and Lamar Counties
- Lavon 345/138-kV substation addition with two 345/138-kV transformers in Collin and Hopkins Counties
- Norwood Switch to Topeka Termination 345-kV line upgrade in Dallas County
- Skyview 345/138-kV substation addition with one 345/138-kV transformer in Dallas County
- Batchler Road to Stainback to Watermill 345-kV double-circuit line upgrades in Dallas County
- Batchler Road to Walker Road 345-kV double-circuit line upgrades in Dallas County
- Watermill Switch to Loop Nine Switch 345-kV line upgrade in Dallas County
- Watermill to Alba Rd to Green Road 345-kV double-circuit line upgrades in Dallas County
- Watermill to Tri Corner 345-kV double-circuit line upgrades in Dallas County
- Tri Corner to Forney Switch 345-kV line upgrade in Dallas and Kaufman Counties
- Trinidad SES to Tri Corner 345-kV double-circuit line upgrades in Dallas and Henderson Counties
- Trinidad SES to Richland Chambers 345-kV line upgrade in Freestone and Henderson Counties
- Ironwood 345-kV substation addition and 345-kV line reconfiguration in Dallas, Ellis, and Navarro Counties
- Notus Switch to Old Country to Venus North 345-kV line upgrades in Ellis and Navarro Counties
- Navarro to Outlaw Switch 345-kV line upgrade in Navarro County
- Sam Switch to Four Brothers Switch 345-kV line upgrade in Hill County
- Tradinghouse SES to Four Brothers 345-kV line upgrade in Hill and McLennan Counties
- Possum Trot Area 345-kV reconfiguration and line upgrades in McLennan, Falls, and Bell Counties
- Possum Trot to Tradinghouse SES to Tiger Creek to Sam Switch 345-kV line upgrades in McLennan and Hill Counties
- Possum Trot to McLennan Data to Lake Hall Switch to Tradinghouse SES 345-kV line upgrades in Hill and McLennan Counties
- Knob Creek Switch to Salado Switch 345-kV line upgrade in Bell County
- Tower Switch area 345-kV reconfiguration and line upgrades in Bell County
- Bell County East Switch to Temple Pecan Creek 345-kV line addition in Bell County

- Bell County East Switch to Littlepond, and Bell County East Switch to Brangus Switch to Gibson 345-kV line upgrades in Bell and Milan Counties
- TNP One Plant Switchyard to Hogcreek to Littlepond 345-kV line upgrades in Milam and Robertson Counties
- Voss Lake 345-kV substation addition with one 345/138-kV transformer and Voss Lake to Bell County East 345-kV line upgrade in Milam County
- Muscovy 345-kV substation addition with two 345/138-kV transformers and Muscovy to Voss Lake 345-kV double-circuit addition in Milam and Williamson Counties
- Big Brown SES to Jewett 345-kV double-circuit line upgrades and 345-kV substation rebuilds in Leon and Limestone Counties
- Jewett to Stranger Switch 345-kV line upgrade in Leon and Limestone Counties
- Monticello to Woodard to Paris Switch 345-kV line upgrades in Titus and Hopkins Counties
- Euclid 765/345-kV substation expansion and Euclid to Hillje 765-kV line addition in Caldwell, Gonzalez, Lavaca, and Wharton Counties
- Hornsby and Gilleland Creek 345-kV continuous voltage support device additions in Travis County
- Dunlap area 345-kV line reconfiguration in Travis County.
- Austrop 345/138-kV transformer upgrades in Travis County.
- Central Texas 345-kV line upgrades in Travis, Bastrop, Caldwell, and Guadalupe Counties.
- Lytton Springs 345/138-kV transformer upgrades in Caldwell County
- Martinez 345/138-kV transformer addition in Bexar County
- Hillje to Crasco Creek to Holman 345-kV line upgrades in Wharton, Caldwell, and Fayette Counties
- South Lane City 345/138-kV substation expansion and South Lane City to Bailey 345-kV double-circuit line addition in Matagorda and Wharton Counties
- Peters area 345/138-kV substation expansion in Fort Bend and Wharton Counties
- PH Robinson to CNP010 to Oasis 345-kV line addition in Brazoria, Galveston, and Harris Counties
- Cachena 345-kV reactor addition in Lavaca County
- Cachena to Elm Creek to Marion 345-kV line upgrades in Lavaca, Wilson, and Guadalupe Counties
- Haber 345-kV capacitor bank addition in Jackson County
- Angstrom to Whitepoint 345-kV line upgrade in San Patricio County
- Calabaza to Whitepoint 345-kV line addition in Nueces and San Patricio Counties
- Nelson Sharpe 345/138-kV transformer addition in Nueces County

- Mantle to Bonilla to Rio Hondo to Gulch to Candela 345-kV line upgrades in Cameron County

The 2025 RTP also included an economic analysis of the ERCOT transmission system for years 2027 and 2030 using both the production cost savings test and the congestion cost savings test. Through this analysis, ERCOT identified transmission congestion and tested various transmission improvements to address the congestion in a cost-effective manner (as defined by ERCOT's economic planning criteria). Specifically, the 2025 RTP evaluated 12 economic transmission improvement projects. Section 4 of this report details the economic analysis results.

The estimated project completion years provided in the 2025 RTP report were selected to address reliability needs in a timely manner. While TOs are expected to meet these project completion dates, actual project timelines may vary due to factors such as construction clearance, regulatory or governmental approval times, equipment availability, land acquisition, and resource constraints.

The projects identified in the 2025 RTP do not represent ERCOT's endorsement of the projects; instead, they represent suggested CAPs or placeholder projects for the reliability criteria violations identified under the system conditions studied in the RTP. The project scopes identified in the RTP may change based on further analysis by ERCOT or the TPs that indicate more cost-effective alternatives or a need to modify the projects due to changes in expected generation, load forecasts, or other system conditions. To confirm need, TPs should perform studies with the latest system conditions and develop applicable reliability projects to resolve any reliability criteria violations.

For projects that are subject to ERCOT Protocols Section 3.11.4, Regional Planning Group Project Review Process, a review shall be conducted in accordance with the process described therein. For a project that is under RPG review when an RTP is developed, a placeholder project will be used if the need is identified. Projects requiring RPG endorsement will be reviewed in future assessments (where sufficient lead time exists), such as future RTPs, to ensure the identified system facilities are still needed.

The TOs will provide ERCOT with additional details on project scope, project cost, and an implementation schedule with completion date(s) for each identified project. This information from the TOs may be provided through further RPG review and/or Transmission Project Information Tracking updates in accordance with ERCOT Planning Guide Section 6.4.1.

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1. 2025 Regional Transmission Plan

This report documents the 2025 Regional Transmission Plan (RTP) assessment performed by ERCOT Grid Planning. It is intended, in part, to satisfy ERCOT's requirements under (North American Electric Reliability Corporation) NERC Reliability Standards, ERCOT Protocols Section 3.11, and ERCOT Planning Guide Sections 3 and 4.

The RTP study is conducted annually for the entire ERCOT System. The 2025 RTP's near-term and long-term planning horizon analysis evaluated the reliability needs of the ERCOT transmission system for the years 2027 through 2031. As required by NERC Reliability Standard TPL-001-5.1, the 2025 RTP included a steady-state analysis of summer peak conditions for years 2027 (year 2), 2028 (year 3), and 2030 (year 5); and off-peak conditions for 2028 (year 3); and a short circuit analysis of summer peak conditions for 2028 (year 3). The 2025 RTP also included steady-state analysis of summer peak conditions for 2031 (year 6), representing the long-term planning horizon. Year six, i.e., 2031, was selected based on the rationale that most transmission upgrades in the ERCOT Region can be completed within five to six years from the date when the need is identified. In addition to analyzing the reliability needs of the system, the 2025 RTP also evaluated economic/efficiency needs of the ERCOT system for years 2027 and 2030.

1.1. Stakeholder Involvement

The development of the RTP is a collaborative process. ERCOT worked with NERC-registered Transmission Planners (TPs), Transmission Owners (TOs), and other stakeholders to develop the input assumptions and scope of technical studies that define the 2025 RTP. These assumptions are described in the RTP Scope and Process document. The RTP Scope and Process document and input assumptions can be found in Appendices A, B, C, and D. Stakeholders were provided with routine updates on the input assumptions and supporting analysis performed for the 2025 RTP at Regional Planning Group (RPG) meetings.

The RPG is responsible for reviewing and providing comments on proposed transmission projects in the ERCOT Region. Under ERCOT Protocols Section 3.11.3, participation in the RPG is required by all Transmission Service Providers (TSPs) and is open to all Market Participants, consumers, other stakeholders, and Public Utility Commission of Texas (PUCT) staff.

ERCOT worked with TPs, TOs, and other stakeholders to study the existing system and identify system upgrades and new transmission projects to ensure continued system reliability.

1.2. Standards and Regulations

The RTP assessment was conducted based on requirements in NERC Reliability Standards, ERCOT Protocols, and the ERCOT Planning Guide.

ERCOT performed its steady-state reliability analysis in accordance with NERC Reliability Standard TPL-001-5.1, Transmission System Planning Performance Requirements. A portion of the RTP

assessment also addressed some requirements in NERC Reliability Standards FAC-002⁸ and IRO-017.⁹

ERCOT Protocols Section 3.10.8.4(3) requires ERCOT to identify additional Transmission Elements that have a high probability of providing significant added economic efficiency to the ERCOT market through the use of Dynamic Ratings and request such Dynamic Ratings from the associated ERCOT TSPs. This report identifies such Transmission Elements as part of its economic analysis.

The RTP assessment adheres to ERCOT Planning Guide Section 3.1.1.2, which provides guidelines regarding completion of the RTP. This section requires that ERCOT complete and publish the final RTP report no later than December 31 each year. Additionally, ERCOT Planning Guide Section 4 and ERCOT Protocols Section 3.11.2 specify the transmission planning criteria to be used in the RTP assessment.

1.3. Confidentiality and Report Posting

The RTP report is shared with internal and external stakeholders. One redacted version of the RTP is created by removing, at a minimum, any confidential data such as the list of long lead-time equipment. The redacted version is shared with ERCOT stakeholders via the MIS Secure Area. A public version of the RTP report is also created by removing, at a minimum, any confidential data and ERCOT Critical Energy Infrastructure Information (ECEII). This report is posted to the ERCOT website.

⁸ FAC-002, Requirement R4

⁹ IRO-017, Requirements R3 and R4

2. 2025 Regional Transmission Plan Process

The RTP study process is illustrated in Figure 1. The initial start cases for the reliability analysis were prepared during the case conditioning stage. For the 2025 RTP, this case conditioning stage included determining appropriate Weather Zone load levels for the RTP study. The details of this methodology can be found in ERCOT Planning Guide Section 3.1.7, Steady State Transmission Planning Load Forecast. In the 2025 RTP, the Permian Basin load forecast from the S&P Global Commodity Insights (formerly IHS Markit) was utilized for the West and Far West Weather Zones. In addition, a significant number of large loads substantiated by signed contracts and offer letters from the TSPs were incorporated in the 2025 RTP.

Following case conditioning, a reliability analysis was conducted on the base cases to determine the Corrective Action Plans (CAPs) needed to meet ERCOT and NERC reliability requirements. In addition to the base cases, the 2025 RTP included sensitivity cases, a short circuit analysis, a cascade analysis, a known outages study, and a multiple element outage analysis as required by NERC Reliability Standard TPL-001-5.1. A minimum deliverability analysis was also performed based on the criteria defined in ERCOT Planning Guide Section 4.1.1.7, Minimum Deliverability Criteria, and the threshold approved by the ERCOT Board of Directors.¹⁰

In the 2024 RTP, a maintenance outage analysis was performed based on the criteria defined in ERCOT Planning Guide Section 4.1.1.8, Maintenance Outage Reliability Criteria. However, the maintenance outage analysis was not performed as a part of the 2025 RTP. The protracted load forecast process, as outlined in Section 2.1, too severely constricted the remaining available timeline to complete all components of the 2025 RTP by the December 31 deadline. On November 12, 2025, ERCOT filed a Good Cause Exception (GCE) with the PUCT¹¹ requesting approval to not perform the Maintenance Outage analysis as required by ERCOT Planning Guide 4.1.1.8. The PUCT approved the request on December 12, 2025.

An economic analysis was also conducted to identify transmission projects that allow reliability criteria to be met at a lower total cost. The detailed scope, process, and input assumptions used in conducting the reliability and economic analyses are available in Appendices A, B, C, and D.

¹⁰ https://www.ercot.com/files/docs/2022/06/28/Minimum_Deliverability_Criteria_Thresholds.pdf

¹¹ https://interchange.puc.texas.gov/Documents/55999_174_1557350.PDF

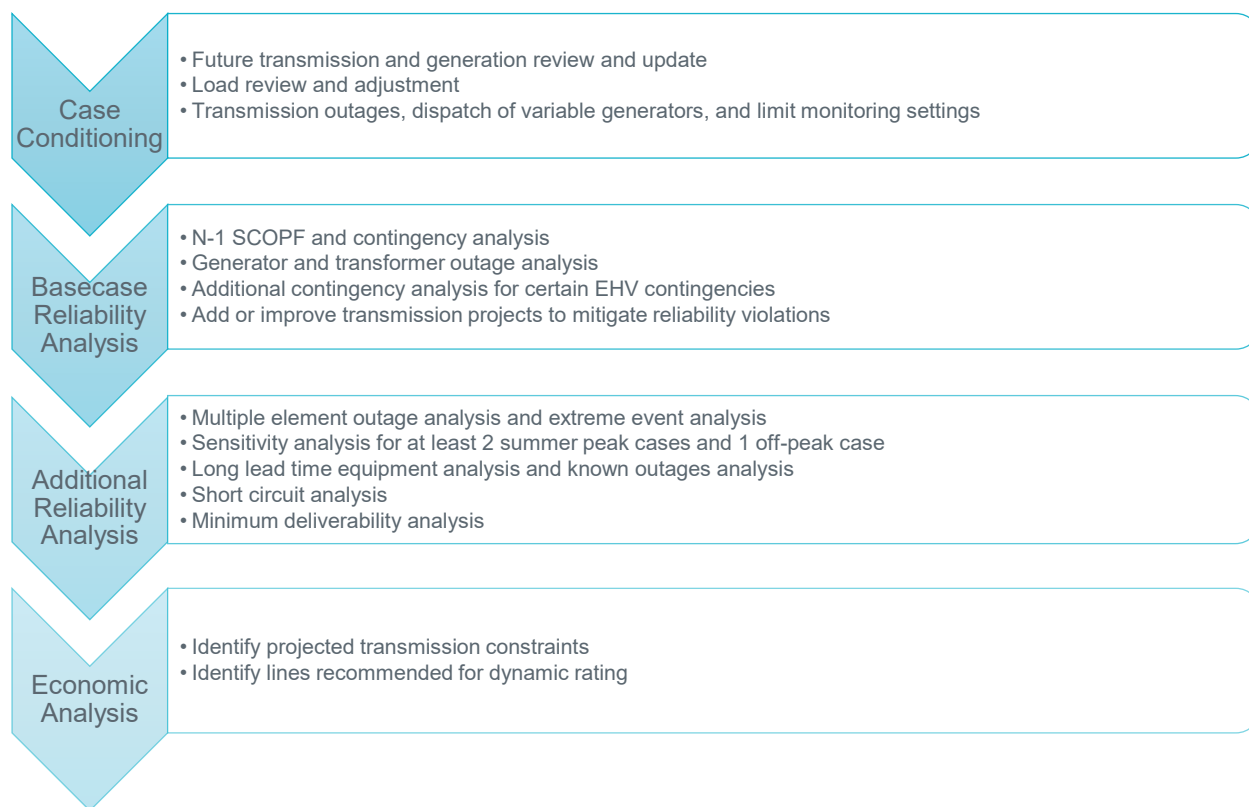


Figure 1: 2025 RTP Transmission Planning Process

ERCOT utilized the following software tools while performing the 2025 RTP:

- PSS/E version 35 was used to develop the conditioned cases.
- PowerWorld version 24 utilized Security Constrained Optimal Power Flow throughout the analysis.
- PowerWorld version 24 was used to screen critical contingencies while evaluating P3 (generator outage) and P6-2 (transformer outage) planning events.
- PowerWorld version 24 was used to perform multiple element outage analysis and cascading analysis.
- UPLAN version 12.3.0.30786 was used to perform security-constrained economic analysis.

2.1. Permian Basin Load Forecast and Large Load Additions

ERCOT and TSPs serving West Texas oil and gas load have been working proactively to better prepare for the transmission planning challenges introduced by the rapid load growth in the Permian Basin and the short lead time of oil and gas load interconnection requests. This effort aims to better understand oil and gas activities and growth, positioning the Texas grid for potential long lead time transmission enhancements needed to reliably serve the fast-growing loads.

In 2019, ERCOT completed the Delaware Basin Load Integration Study with extensive input from TSPs, identifying a five-stage transmission upgrade road map to reliably serve different levels of

Delaware Basin load. In 2021, ERCOT completed the Permian Basin Load Interconnection Study, informed by the TSP-sponsored IHS Markit study report for the Permian Basin load forecast, which provided a detailed load forecast based on an in-depth analysis of oil and gas activities in the Permian Basin. In 2022, S&P Global Commodity Insights refreshed the Permian Basin load forecast, showing continued load growth and grid connection needs. The load forecast from 2027 to 2040 in the Permian Basin area is shown in Figure 2. In July 2024, as directed by the PUCT, ERCOT developed the Permian Basin Reliability Plan which incorporated the S&P Global Commodity Insights Permian Basin load forecast. This plan contains both local upgrades and new import paths to address reliability issues in the Permian Basin area. The refreshed S&P Global Commodity Insights Permian Basin load forecast was incorporated in the 2024 and 2025 RTPs.

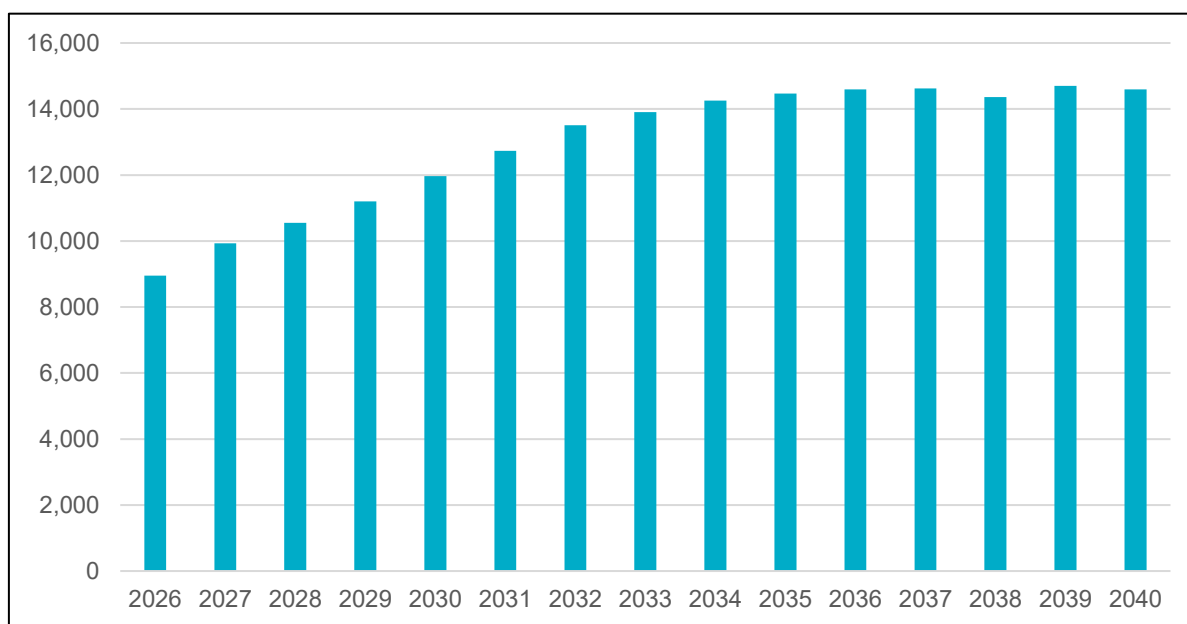


Figure 2: S&P Global Commodity Insights Permian Basin Summer Peak Load Forecast (MW)

As the new load interconnection timeline outpaces the development of transmission projects, the Texas Legislature recognized the need to update rules in the ERCOT Region. Previously, these rules did not allow ERCOT to include loads in planning studies without signed interconnection agreements. This resulted in the inclusion of approximately eight gigawatts (GW) of large loads in both the 2022 and 2023 RTPs.

In December 2022, the Public Utility Commission of Texas (PUCT) amended 16 Texas Administrative Code (TAC) § 25.101, Certification Criteria, to implement Senate Bill 1281 (87th Legislative Session). The amendment included the requirement for any review conducted by ERCOT to incorporate historical load, forecasted load growth, and additional load currently seeking interconnection.

House Bill (HB) 5066 (88th Legislative Session) introduced a new requirement to consider loads for which the electric utility has yet to sign an interconnection agreement, as determined by the electric utility responsible for serving the load. Consequently, ERCOT accepted attestation letters from TSP

officers (“Officer Letters”) as substantiation for inclusion in the analysis. These new requirements were first implemented in the 2024 RTP and led to a significant increase in large loads included in that assessment (i.e., data centers, crypto mining, hydrogen and hydrogen-related manufacturing, and electrification). In the 2024 RTP, the forecasted summer peak demand for 2030 exceeded 150 gigawatts (GW), with close to 40 GW attributed to large loads seeking interconnections under the new requirements.

The large loads submitted by TSPs for consideration under the HB5066 requirements in the 2025 RTP saw an even more significant increase compared to the 2024 RTP. The forecasted summer peak demand for 2031 would have exceeded 218 GW if all TSP-submitted large loads were included. After performing historical analysis on large load realization, ERCOT developed an ERCOT Transmission Planning Adjusted Load Forecast (ETPALF) which adjusted the 2031 forecast to 159 GW. The ETPALF preserved large loads substantiated with an interconnection agreement at 100%; large loads substantiated with officer attestations (“Officer Letter Loads” – OLLs) were reduced to 55.4%; and, data center OLLs were further reduced to 49.8% of the initial OLL reduction, resulting in a final amount that was 27.6% of the full data center value. The ETPALF methodology resulted in the incorporation of over 60 GW of large loads in the 2025 RTP. ERCOT requested a Good Cause Exception (GCE) from the PUCT to use the ETPALF in the 2025 RTP. The GCE was approved on June 5, 2025. The solid green line in Figure 3 below shows the 2025 RTP system load levels that resulted from the ETPALF methodology approved in the GCE.

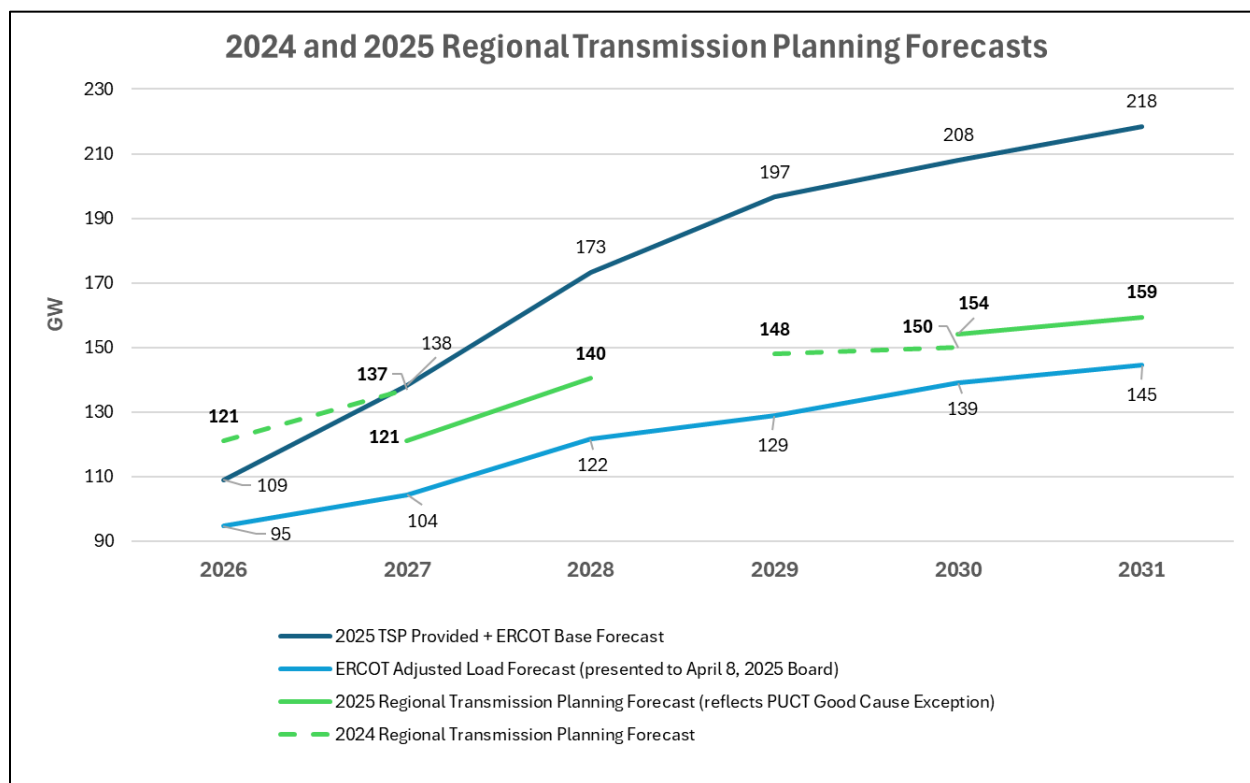


Figure 3: RTP System Load Comparison

Figure 4 shows the amount of large load included in each study year. These large loads include data center load, cryptocurrency load, and hydrogen and hydrogen-related manufacturing. The breakdown of different load types is shown in Figure 5.

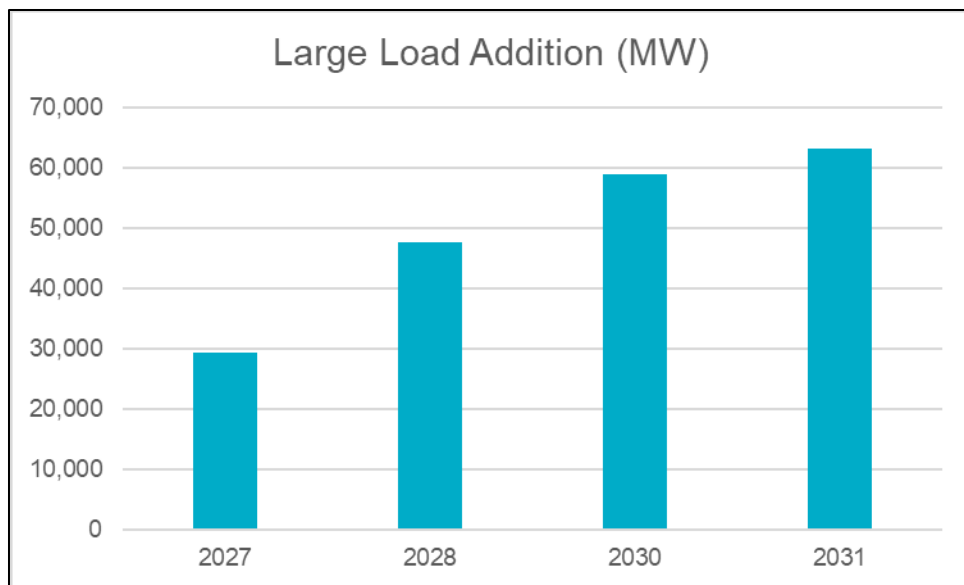


Figure 4: Large Load Addition in 2025 RTP (MW)

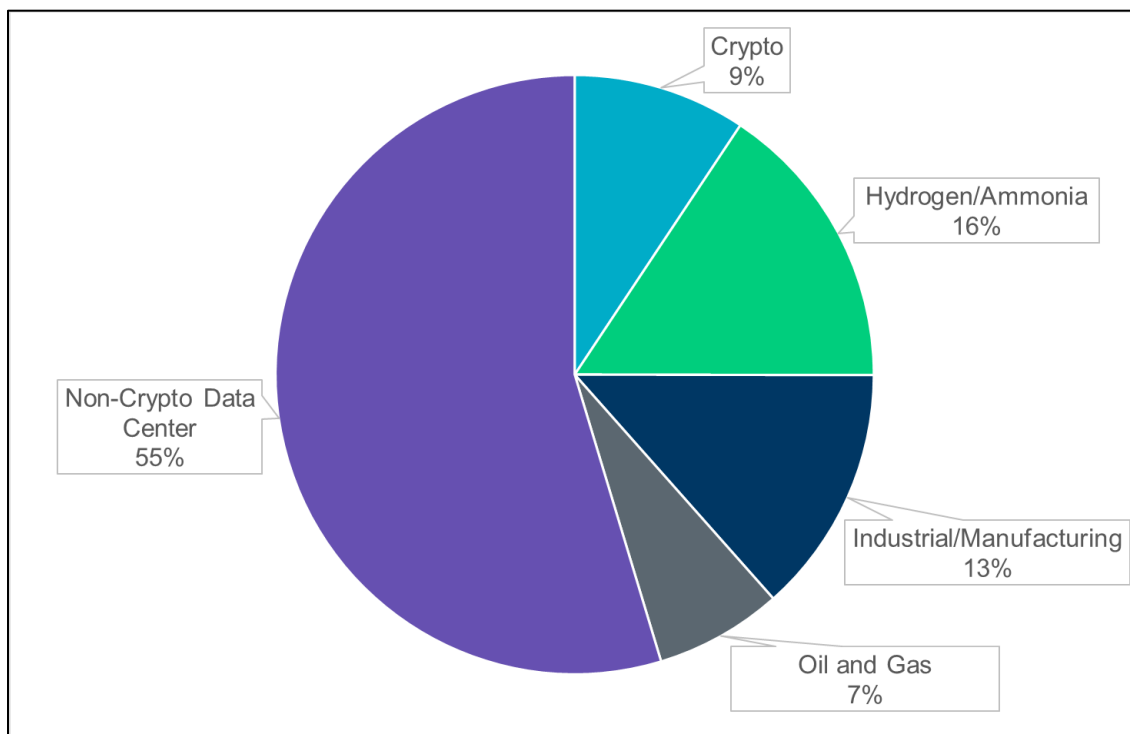


Figure 5: Types of Large Loads in 2025 RTP

Another critical characteristic associated with large load interconnection requests is the high concentration of requests in certain areas of the system. This created additional challenges in

transmission planning, including both import needs and local transmission needs. Figure 6 shows the location of the large loads in the 2025 RTP. The load values used in Figure 6 represent the adjusted value based on the ETPALF methodology.

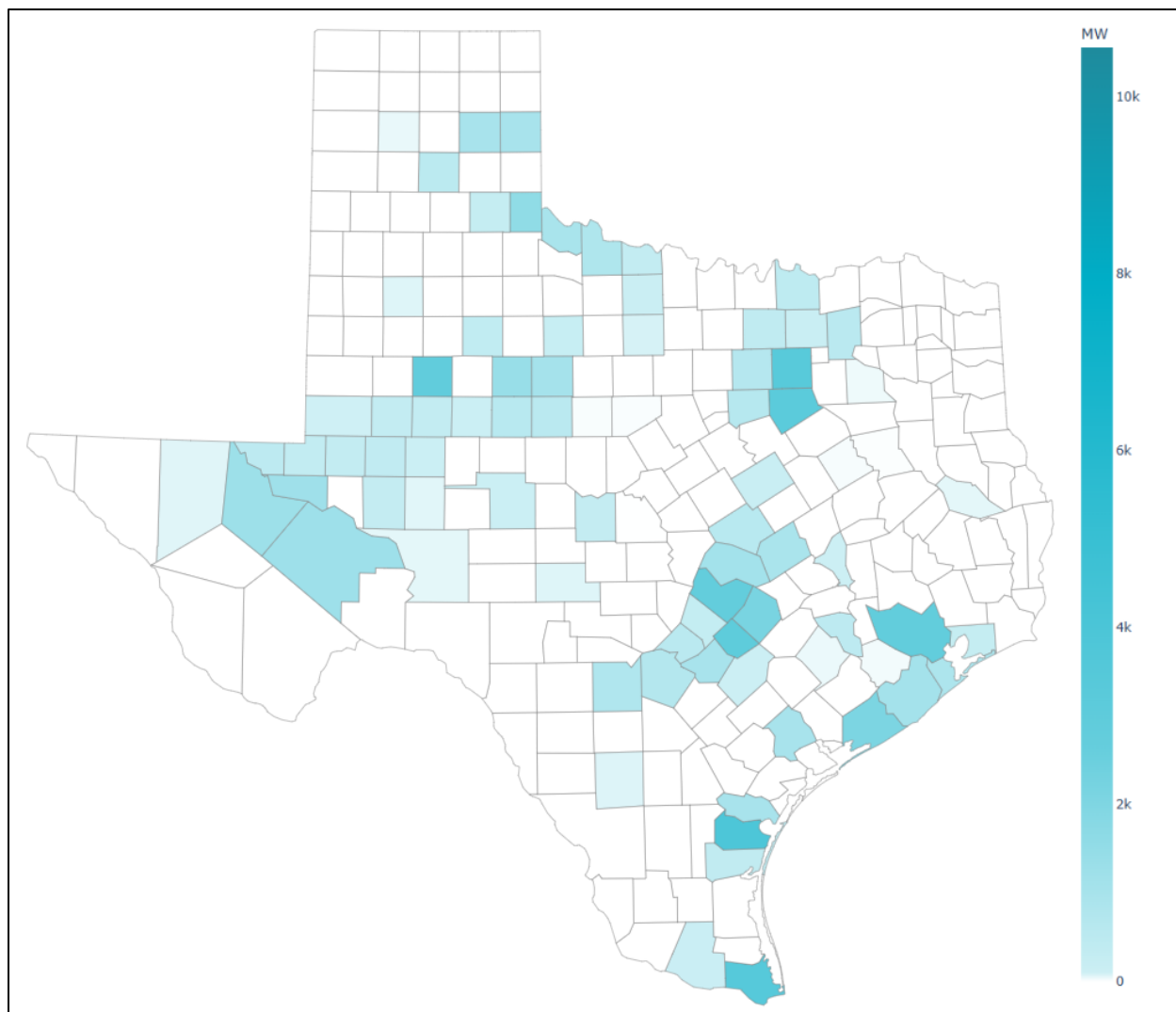


Figure 6: Location of Large Loads in 2025 RTP

2.2. Adoption of the ERCOT Rooftop Solar Growth and Electric Vehicle (EV) Load Impact Forecast

The rapid growth in Distributed Generation (DG), especially in the solar photovoltaic less than 1 megawatt (MW) category, continued in the ERCOT Region. The total DG at the end of 2024 was estimated to be more than 6,000 MW, as shown in Figure 7, which represents an approximately 27% increase from 2023.

Similar to the previous three RTPs, the impacts of the projected rooftop solar growth were incorporated as load reductions at the bus level in the 2025 RTP.

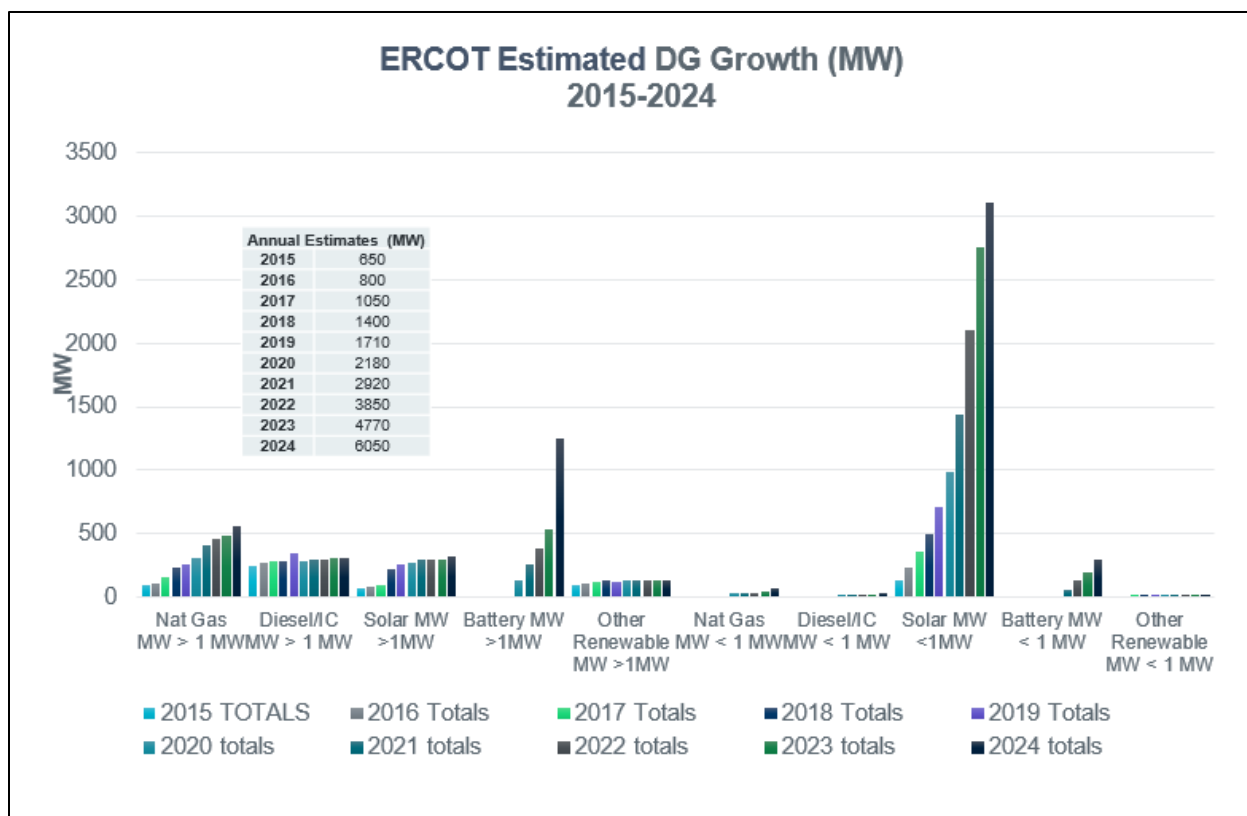


Figure 7: ERCOT Estimated Total DG Growth from 2015 to 2024 (MW)

Adoption of EVs is expected to increase significantly in the near future, which signifies a need to include EV load impacts in near-term planning studies. Similar to the 2023 and 2024 RTPs, the 2025 RTP incorporated the substation-level EV load impacts based on the methodology and process¹² developed by the Brattle Group.

2.3. Generation Assumptions

ERCOT took steps outside of the normal planning process to incorporate more generation into the planning models used in the 2025 RTP reliability analysis based on the rules proposed by Planning Guide Revision Request (PGRR) 127¹³. PGRR127 modifies the criteria for adding Generation Resources to the planning models and is currently going through the stakeholder process. The generators were added in the sequence outlined below.

1. Added all units that met Planning Guide 6.9(1) requirements
2. Added all units with a signed Interconnection Agreement (IA)
3. Added all units with a completed Full Interconnection Study (FIS)
4. Added all units that had started the FIS and were active participants of the Texas Energy Fund (TEF)

¹² <https://www.ercot.com/files/docs/2023/08/28/ERCOT-EV-Adoption-Final-Report.pdf>

¹³ <https://www.ercot.com/mktrules/issues/PGRR127>

Figure 8 shows the generation categories added to each study base case to serve load and losses and provide approximately 3,000 MW of generation reserves.

2027 SUM	2028 MIN & SUM	2030 SUM	2031 SUM
6.9(1)	6.9(1) IA	6.9(1) IA FIS Completed	6.9(1) IA FIS Completed FIS Started – TEF

Figure 8: Generation Categories Added to Each Study Base Case

2.4. Reliability Analysis

The reliability analysis in the 2025 RTP was focused on the steady-state analysis requirements of NERC Reliability Standard TPL-001-5.1 and the ERCOT Planning Guide. The purpose of the reliability analysis was to identify potential criteria violations and CAPs that may be used to resolve them. The RTP analysis included Security Constrained Optimal Power Flow (SCOPF) to identify unresolvable constraints. Loading and voltage levels of Bulk Electric System (BES) elements were monitored for all NERC planning events, including extreme events. ERCOT staff developed CAPs in collaboration with TPs to mitigate criteria violations in accordance with the NERC and ERCOT performance requirements.

The 2025 RTP reliability analysis included the following studies:

- SCOPF: Security Constrained Optimal Power Flow (SCOPF) was used to perform basic power flow and contingency analysis for P0, P1, P2-1, and P7 planning events. SCOPF used generation cost data and other system constraints to give an optimal generation dispatch and unit commitment while maintaining the reliability of the system. In this analysis, the software simulated the removal of all elements of the Protection System and other automatic controls following the contingency event.
- Contingency Analysis: Basic contingency analysis routines in the power flow software were used to test P2-2, P2-3, P2-4, P4, and P5 planning events and extreme events.
- Multiple Element Contingency Analysis: Planning events P3 and P6 involved a first- and second-level contingency analysis. Such events were tested using multiple element contingency analysis. During this analysis, loss of elements due to the first contingency was followed by acceptable system adjustments before testing the effect of the second contingency event. The list of acceptable system adjustments included system reconfiguration, changes in voltage schedule, and re-dispatch of generation. Other contingency events such as P4 and P5 planning events and extreme events, which involved simultaneous removal of multiple elements, were also analyzed. Extreme events associated with the disruption of gas pipelines were also included.
- Cascading Analysis: Cascading analysis was conducted to test all planning and extreme events where a facility may be loaded above its relay loadability rating before mitigation

measures can be taken. In this analysis, the software simulated the removal of all elements of Protection System and other automatic controls following the contingency event. This included tripping of generators and transmission elements which were loaded beyond their relay loadability limits. These contingencies were screened to detect potential cascade events for more detailed analysis.

- **Short Circuit Analysis:** In accordance with the agreement between ERCOT and TPs in the ERCOT Region as required by NERC Reliability Standard TPL-001-5.1, Requirement R7 (revised in May 2023), ERCOT performed the short circuit analysis to determine short circuit currents for Resource Entity (RE)-owned facilities. The results of the short circuit analysis included the magnitude of short circuit current and the source impedance associated with each fault. These results were communicated to the NERC-registered Generator Owners (GOs). GOs completed a review of study results, acknowledged the findings, and provided a list of over-dutied circuit breakers and CAPs. In addition, GOs also confirmed the continued validity and implementation status of the facilities identified in the previous RTP.
- **Long Lead Time Equipment Analysis:** Under Requirement 2.1.5 of NERC Reliability Standard TPL-001-5.1, the impact of the possible unavailability of major transmission equipment with a lead time of one year or more was studied. The studies were performed with an initial condition of the identified long lead time equipment modeled as out of service, then evaluated under P0, P1, and P2 planning events. The list of long lead time equipment was developed based on feedback from the TOs. The results of this analysis were communicated to the TOs.
- **Sensitivity Analysis:** ERCOT selected the summer peak conditions of 2027 and 2030 and off-peak conditions of 2028 for sensitivity analyses as required by Requirement 2.1.3 of NERC Reliability Standard TPL-001-5.1. ERCOT prepared the following sensitivity cases by varying the generation and load input assumptions:
 - The large loads substantiated by TSP officer letters were reduced to 0 MW.
 - Generators that had not yet met ERCOT Planning Guide 6.9(1) requirements were removed.

The sensitivity analyses were performed using the same start cases as the 2025 RTP base case analysis.

- **Known Outages Impact Analysis:** Under Requirement 2.1.4 of NERC Reliability Standard TPL-001-5.1, the impact of known outages of generation or transmission facilities planned in the near-term planning horizon was studied. ERCOT issued Market Notices to collect known outages information from both TOs and GOs. TOs were required to provide technical rationales for their known outages selection for each study case included in the 2025 RTP. ERCOT developed the technical rationale¹⁴ for the selection of GO-submitted outages to be included in the corresponding study cases. The known outages from the TOs and the GO

¹⁴https://www.ercot.com/files/docs/2022/03/09/2022_RTP_TPL_001-5_Known_Outages_March_2022_RPG.pdf

outages selected based on ERCOT-developed technical rationale were then used to study their impact on system performance under P0 and P1 contingencies.

- **Minimum Deliverability Analysis:** As required by ERCOT Planning Guide Section 4.1.1.7, Minimum Deliverability Criteria, ERCOT performed an analysis to ensure the deliverability of 100% of capacity of Generation Resources utilizing combined cycle, steam turbine, combustion turbine, hydro, or reciprocating engine technology, and for any Energy Storage Resource (ESR) with a duration greater than or equal to two hours. For ESRs with a duration less than two hours, prorated deliverability was ensured. CAPs were proposed to address any reliability violations under the contingencies defined for the minimum deliverability criteria. See Section 3.3 for additional analysis details.

2.4.1. CAP Development

Under the ERCOT Planning Guide, reliability projects are those system improvements that are needed to meet NERC Reliability Standards or ERCOT planning criteria, which could not otherwise be met by simultaneously feasible, security-constrained re-dispatch of existing and planned generation. To develop this list of reliability projects, grid simulation software was utilized, which included the removal of all Protection System elements and other automatic controls following the simulated contingency events. These elements included devices designed to provide steady-state control of electrical system quantities, such as on-load tap-changing transformers, phase-shifting transformers, and switched capacitors and reactors.

A list of potential CAPs, or reliability projects, along with the corresponding limiting elements and contingencies, was communicated to the appropriate TP and/or TO. TPs and TOs reviewed the initial list of reliability-driven projects for their technical feasibility and estimated year of completion (considering necessary lead times). In some cases, the TOs also provided project alternatives. In instances where it is not feasible to construct a project prior to the identified date of need, ERCOT designed Constraint Management Plans (CMPs) to mitigate the criteria violations until the permanent CAP can be put in-service. These mitigation actions were developed in collaboration with TPs and further communicated to ERCOT Operations. The results were posted on the ERCOT MIS Secure Area. Study findings were presented to stakeholders at regularly scheduled RPG meetings to solicit comments and suggestions.

2.5. Economic Analysis

ERCOT conducted an economic analysis to identify system improvements that allow ERCOT to meet NERC Reliability Standards and ERCOT planning criteria more economically than the continued dispatch of higher cost generation.

To identify such economically driven projects, ERCOT created production cost simulation models for years 2027 and 2030. Details on the production cost models developed for the 2025 RTP can be found in Appendices C and D.

According to the economic planning criteria described in ERCOT Nodal Protocols Section 3.11.2, both the production cost savings test and congestion cost savings test, which is measured by the system-wide consumer energy cost reduction, were performed. ERCOT recommends an economic project if:

- the levelized ERCOT-wide annual production cost savings equals or exceeds the first-year annual revenue requirement of the transmission project; or
- the levelized system-wide consumer energy cost reduction equals or exceeds the average of the first three years' annual revenue requirement for the project.

Based on the recent review, the first-year annual revenue requirement for a project was determined to be 13.0% of the estimated project cost, and the average of the first three-year annual revenue requirement for a project was 12.7% of the estimated project cost.

3. Findings from Reliability Analysis

3.1. Base Analysis Reliability Projects and Constraint Management Plans

The primary purpose of the 2025 RTP reliability analysis was to identify reliability criteria violations and potential CAPs to resolve them. Overall, the base reliability analysis identified a need for 192 CAPs. The detailed list of criteria violations and resulting CAPs can be found in Appendix E. Figure 9 illustrates the geographic location of the identified CAPs. The legend linking reliability projects and their associated map indices can be found in Appendix F.

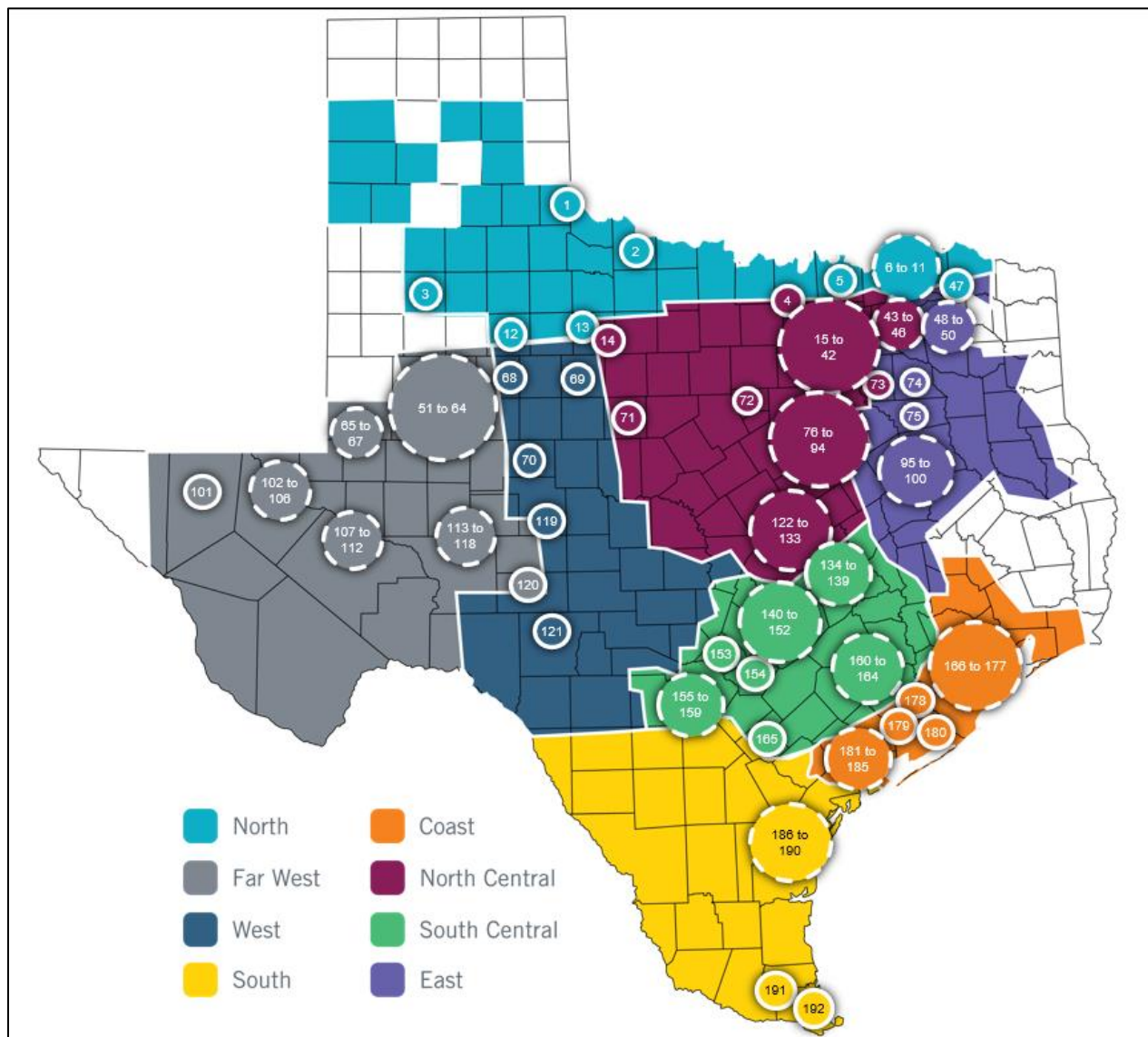


Figure 9: Geographic Locations of CAPs Identified in the 2025 RTP

Figures 10¹⁵ and 11 summarize the types of projects, their geographic locations, and associated voltage levels. Figure 12 distinguishes between projects that were newly identified in the 2025 RTP and projects that were identified in previous ERCOT planning studies or TSP studies.

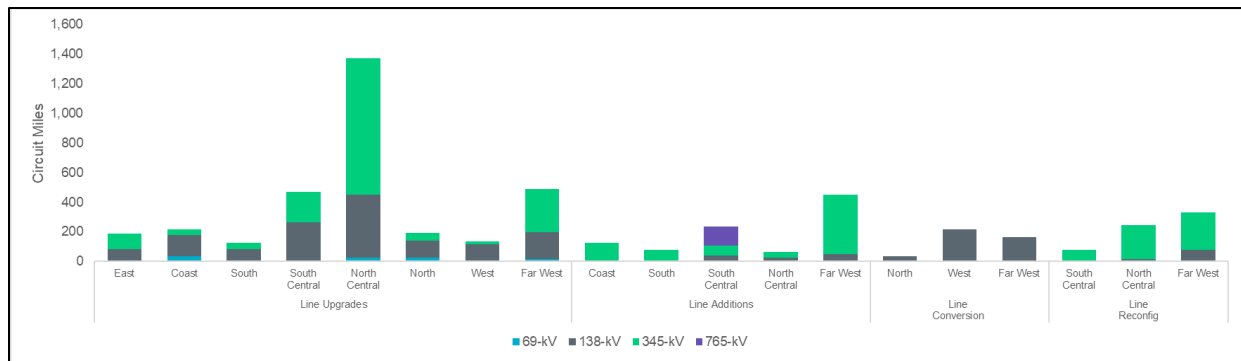


Figure 10: 2025 RTP Transmission Line Project Types by Weather Zone and Voltage Level

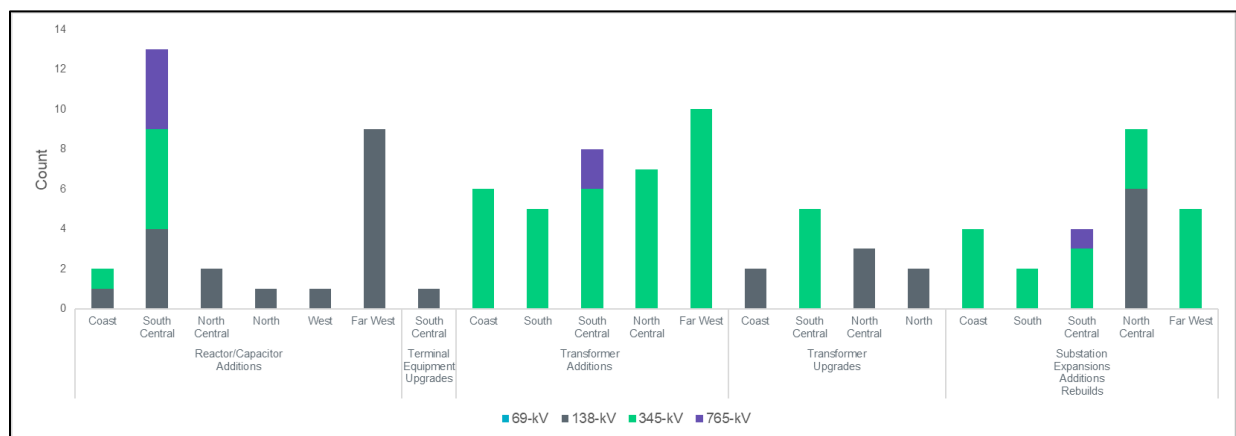


Figure 11: 2025 RTP Other Upgrades and Additions by Weather Zone and Voltage Level

¹⁵ The 69-kV to 138-kV line conversion was included in the 138-kV category.

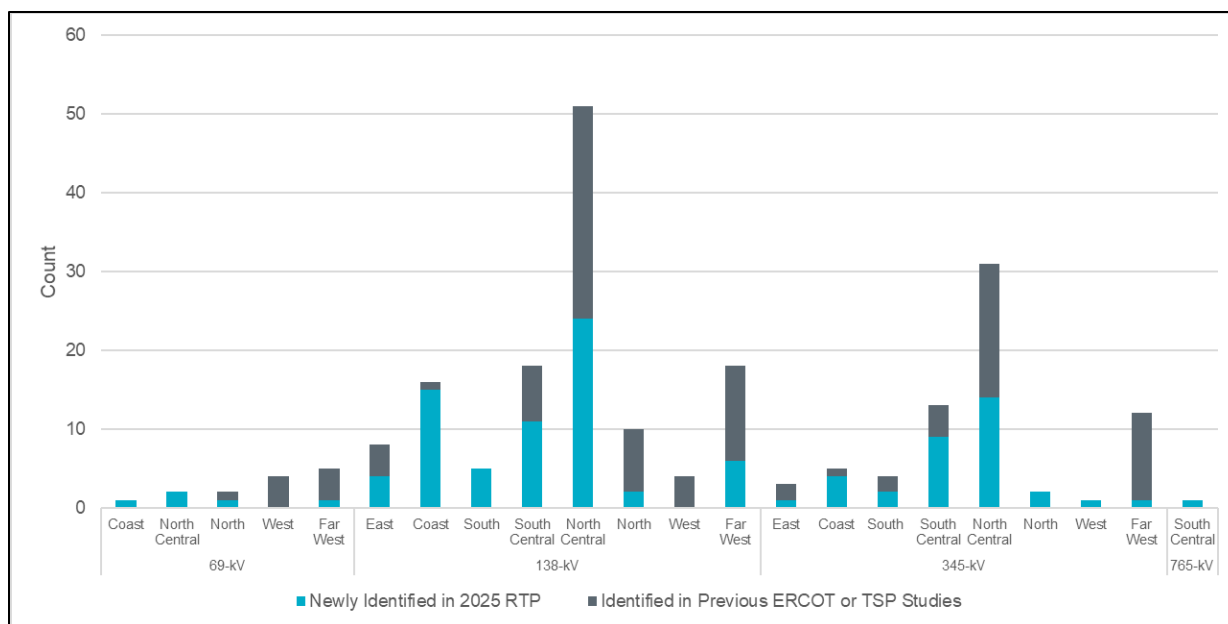


Figure 12: Projects Newly Identified in the 2025 RTP versus Projects Previously Identified

ERCOT, in collaboration with TPs, also identified 11 potential CMPs as placeholder mitigating actions, which will be reviewed in the operations planning horizon by ERCOT and TOs. The list and details of the CMPs identified in the 2025 RTP can be found in Appendix G.

3.1.1 Texas 765-kV (kilovolt) Strategic Transmission Expansion Plan (STEP)

The unprecedented load growth has introduced significant challenges to the ERCOT grid, marking a new era of planning. It is crucial to review and update current planning processes to adapt to these changes. In 2024, ERCOT began exploring the potential introduction of a 765-kV voltage class to the ERCOT Region. This higher voltage class is expected to provide a long-term solution to accommodate the rapid evolution of the ERCOT System. The benefits of higher voltage classes include the following:

- Increased transfer capability to load centers
- Flexibility in Generation Resource siting
- Increase outage coordination capacity
- Reduced impact to Texas consumers due to less overall Right-of-Way requirements¹⁶
- Reduced line losses
- Potential retirement of series compensation devices
- Potential exit strategy for some current Generic Transmission Constraints

¹⁶ A single 765-kV circuit requires approximately 200 feet of Right-of-Way but carries up to three times the power of a 345-kV double circuit. A single 345-kV double circuit requires approximately 150 feet of Right-of-Way. When comparing overall Right-of-Way needs, the 765-kV option requires less than half (200 feet) what the equivalent 345-kV option requires (150 feet x 3 double circuits = 450 total feet for equivalent 345-kV transmission capacity).

In the 2024 RTP, ERCOT developed a 765-kV Strategic Transmission Expansion Plan (STEP) needed by 2030. It compared the 765-kV plan to an alternative 345-kV plan and presented the findings from that comparison at the EHV 765-kV ERCOT Workshop¹⁷ on January 27, 2025. After reviewing the comparison and holding additional public stakeholder discussions, the PUCT approved the 765-kV Permian Basin import paths over the 345-kV option on April 24. On July 7, the Drill Hole to Sand Lake to Solstice 765-kV Line Project, designed to close the western end of the Permian Basin import lines, was submitted to the RPG for ERCOT Independent Review. On July 17, the 765-kV-STEP Eastern Backbone Project was also submitted to RPG for ERCOT Independent Review. ERCOT presented its final recommendations for the projects at the November 11 RPG meeting, and the Technical Advisory Committee approved the projects on November 19. The ERCOT Board of Directors approved both 765-kV RPG projects at the December 2025 Board meeting, thereby concluding the formal approval process for the remaining components of the 765-kV STEP as shown in Figure 13 below.

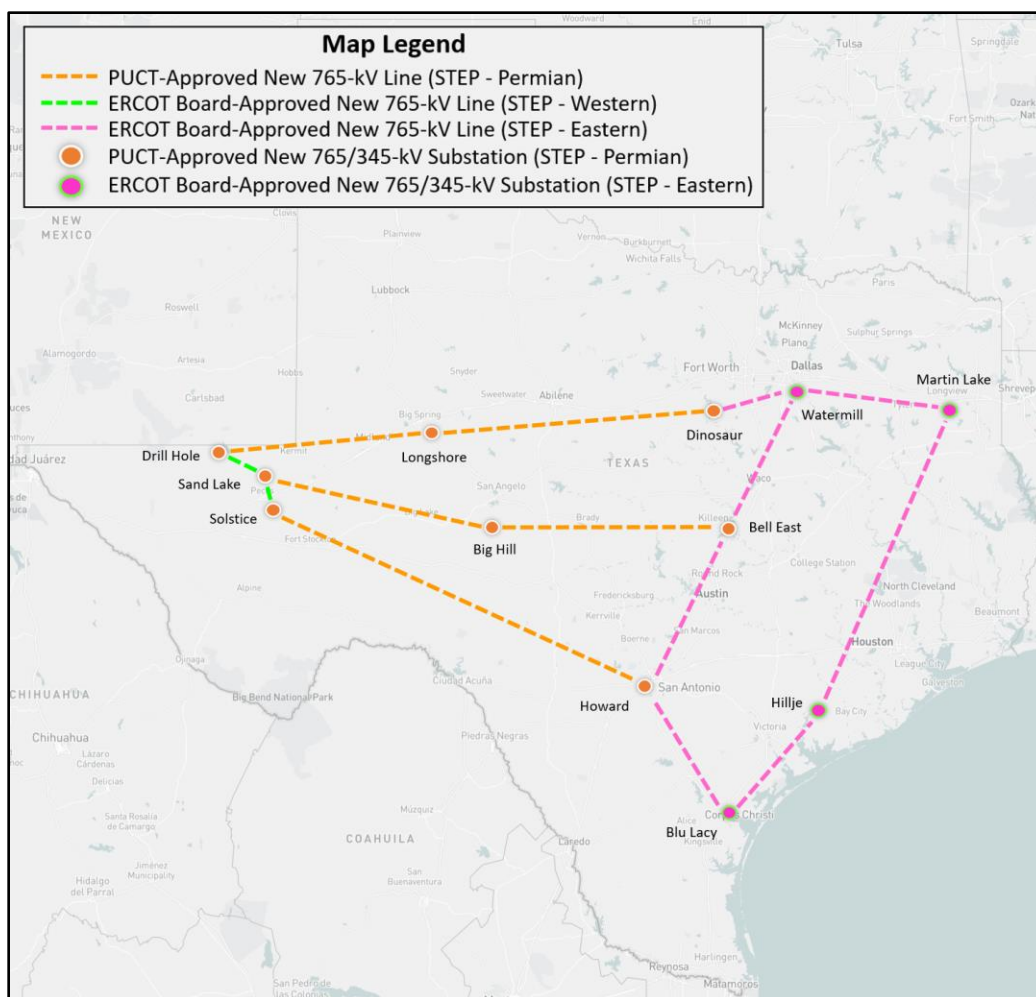


Figure 13: 765-kV Strategic Transmission Expansion Plan (STEP). All components are now PUCT or ERCOT Board of Directors approved.

¹⁷ <https://www.ercot.com/calendar/01272025-EHV-765-kV-ERCOT>

The 2025 RTP included the entire 765-kV STEP in its 2030 and 2031 starting base cases to address base case violations and found it provided continued benefits in addressing new load growth. The 2025 RTP did not see a need to change or remove any element of the 765-kV STEP.

3.1.2 West Texas Study Findings

As described in Section 2.1, the Permian Basin load forecast from the S&P Global Commodity Insights study was adopted in the 2025 RTP, similar to the 2021-2024 RTPs. This projected oil and gas load, along with large loads seeking interconnections, increased the Far West load to nearly 19 GW by 2030 and nearly 20 GW by 2031 under summer peak conditions. In comparison, the 2029 summer peak load forecast for the Far West Weather Zone was under 15 GW in the 2023 RTP. The 2025 RTP incorporated the Permian Basin Reliability Plan local projects needed by 2030 that were approved by the PUCT in September 2024. The 2025 RTP also included the three 765-kV import paths identified by the Permian Basin Reliability Plan and approved by the PUCT in April 2025. The full Permian Basin Reliability Plan, both local projects and import paths, were included in the 2030 and 2031 summer peak cases.

3.1.3 Central Texas Study Findings

It was observed in the 2023 RTP that conventional Generation Resource retirements in and around the San Antonio area contributed to stress on the existing 345-kV Central Texas corridor as new renewable Generation Resources in the South Weather Zone made their way up to support load growth in the South Central Weather Zone. In the 2024 RTP, generation in both the South and South Central Weather Zones increased compared to the 2023 RTP cycle.

Despite generation increases in the South and South Central Weather Zones, the increase in large load interconnection requests outpaced the generation growth. The new large load pockets in the South and South Central Weather Zones have changed the flow patterns from what was observed in the 2023 RTP. The trends seen in the 2024 RTP continued in the 2025 RTP. There was an addition of approximately 12.8 GW of load in the 2031 summer peak case in the Central Texas corridor counties of Bastrop, Bell, Bexar, Caldwell, Comal, Gonzales, Guadalupe, Hays, Medina, Milam, Travis, and Williamson. The approximate large load concentration by county is illustrated in Figure 14.

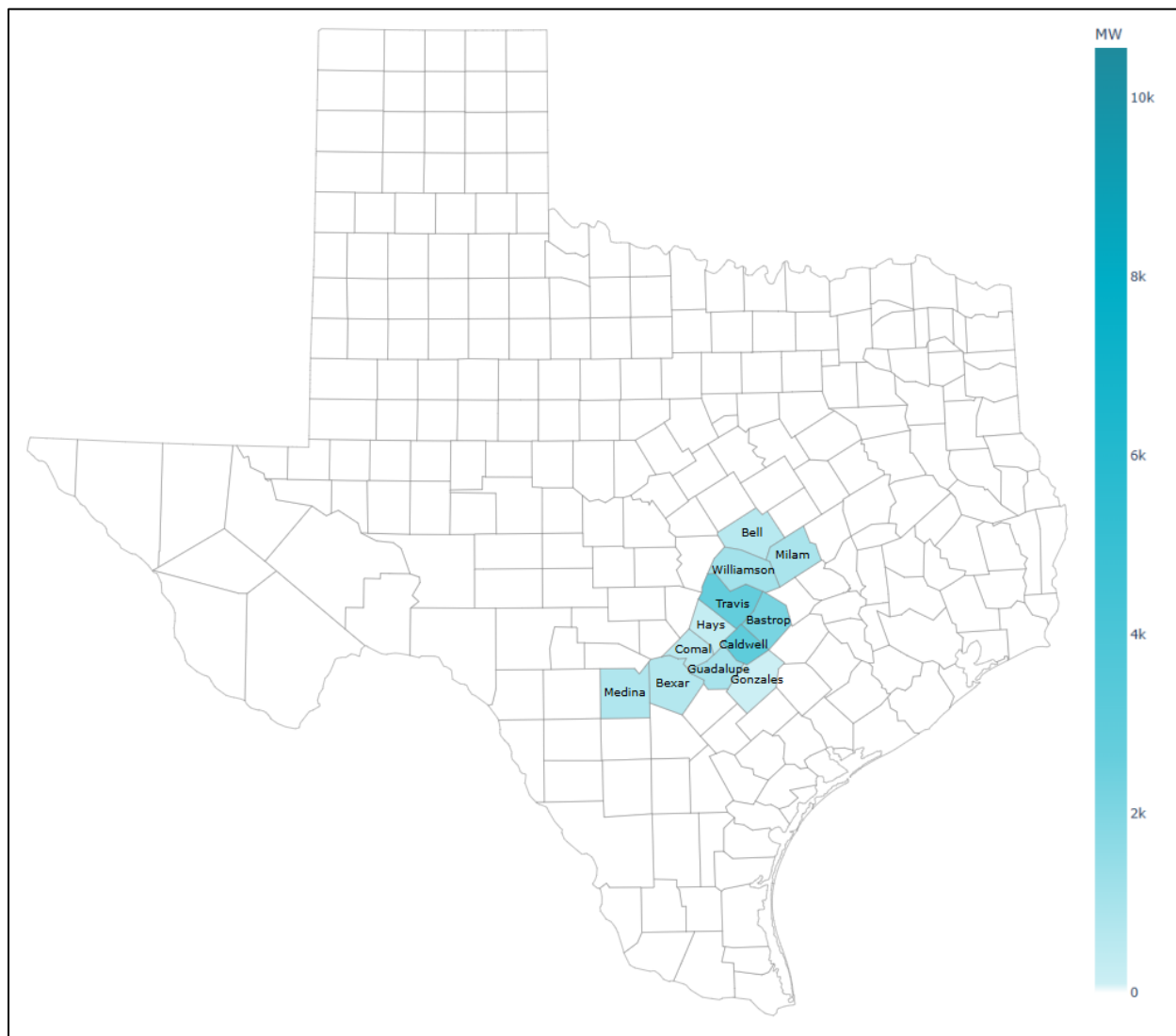


Figure 14: Central Texas Load Pocket Breakdown by County

Renewable resources in the South Weather Zone are now exporting less power to the San Antonio and Austin areas with the increase of large loads in Calhoun, Nueces, and Cameron counties. However, Central Texas faces the challenge of needing additional import capability. To meet the higher projected load demand in Central Texas, new flow patterns have emerged, with imports to Central Texas coming from north of Austin and the Coast Weather Zone.

Like the 2024 RTP, the 2025 RTP showed a need for additional import capacity along the pathway from north of Austin to Central Texas as illustrated by Figure 15. A combination of upgrades along the north-south corridor and new east-west 345-kV lines connecting Sandow Switch to Hutto are needed to resolve the reliability violations in the northern area of Central Texas. A need to have more 345/138-kV transformer capacity in Central Texas to serve the load at the 138-kV voltage level was also observed in both the 2024 and 2025 RTPs.

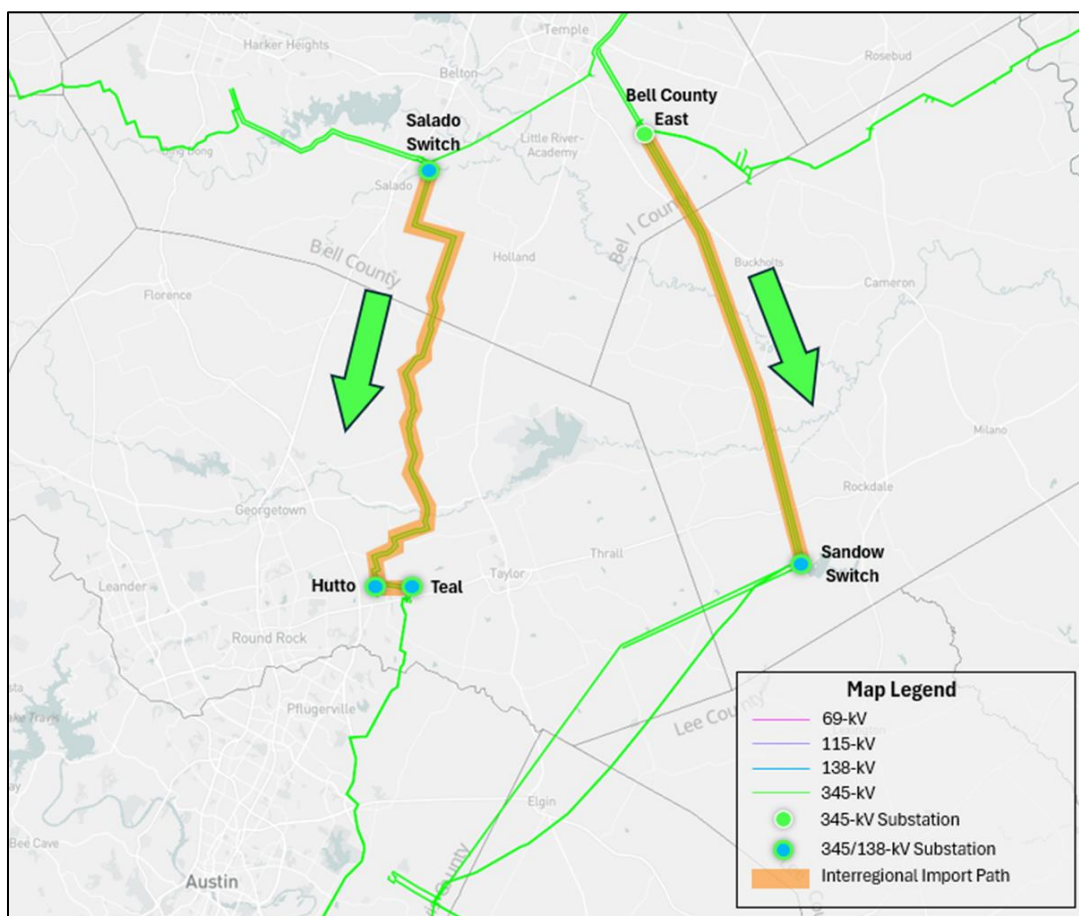


Figure 15: Central Texas Import Path from North of Austin

The reinforcement and expansion of the northern Central Texas import corridor alone is not sufficient to address the issues identified in Central Texas. Strained import capability into the area was first identified in the 2023 RTP, when it developed a South to Central Texas 345-kV import path extending from the Corpus Christi area into Central Texas. However, with the dramatic increase in loads in the Corpus Christi area in the 2024 RTP, there was less generation available for export. Instead, the 2024 RTP reinforced the existing 345-kV import corridors from Coast to South Central. The two import paths coming from the Coast Weather Zone to Central Texas currently consist of lower-rated 345-kV lines which limit the power transfer capability into the area to a level below what's needed. These two paths are illustrated in Figure 16.

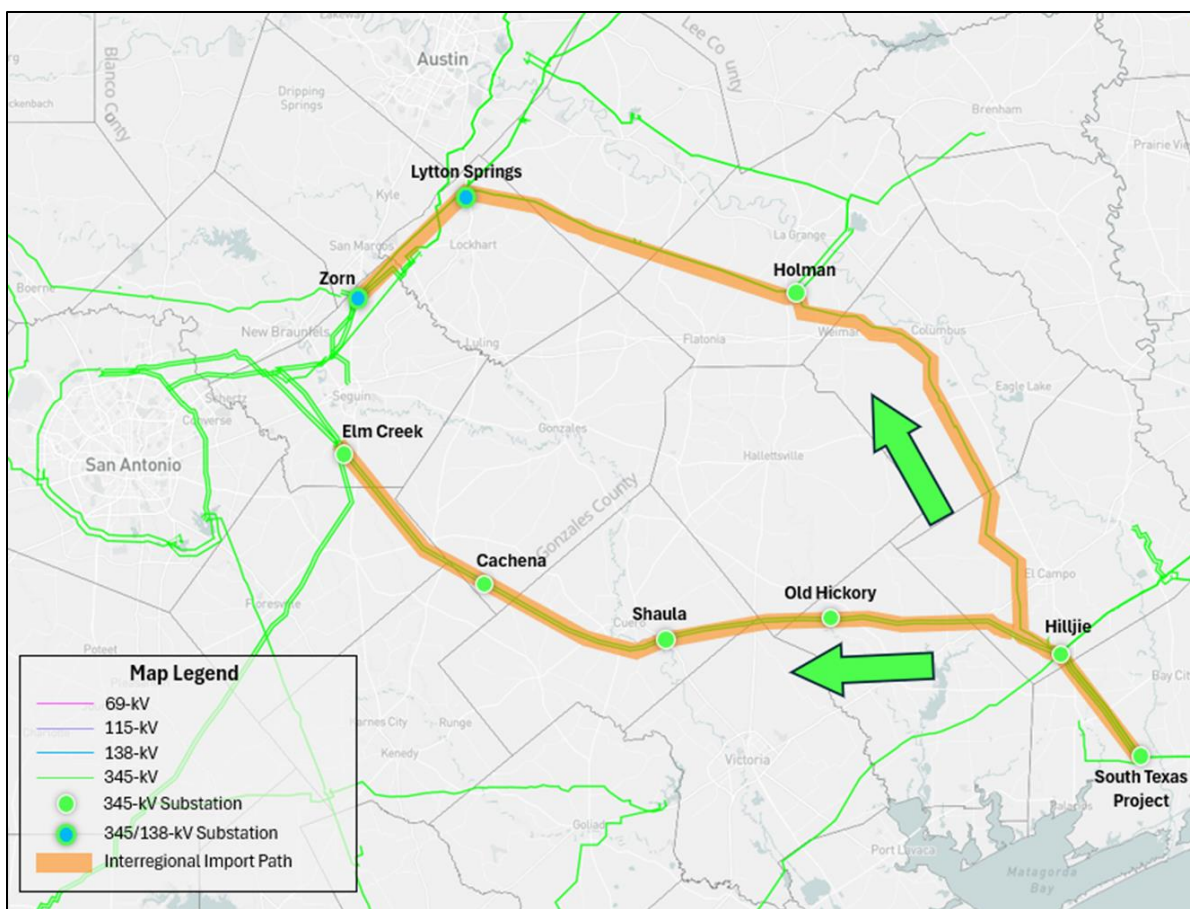


Figure 16: Coast to South Central Import Path

Given the amount of current and future generation additions along the already-constrained Coast-to-Central Texas import paths, upgrades to those existing pathways are also needed under the system conditions seen in the 2025 RTP. Upgrades of the existing 345-kV pathways are also required; however, those are considered complimentary components of the overall Central Texas import strategy while the foundational piece is a new 765-kV import path into the area as shown in teal in Figure 17.

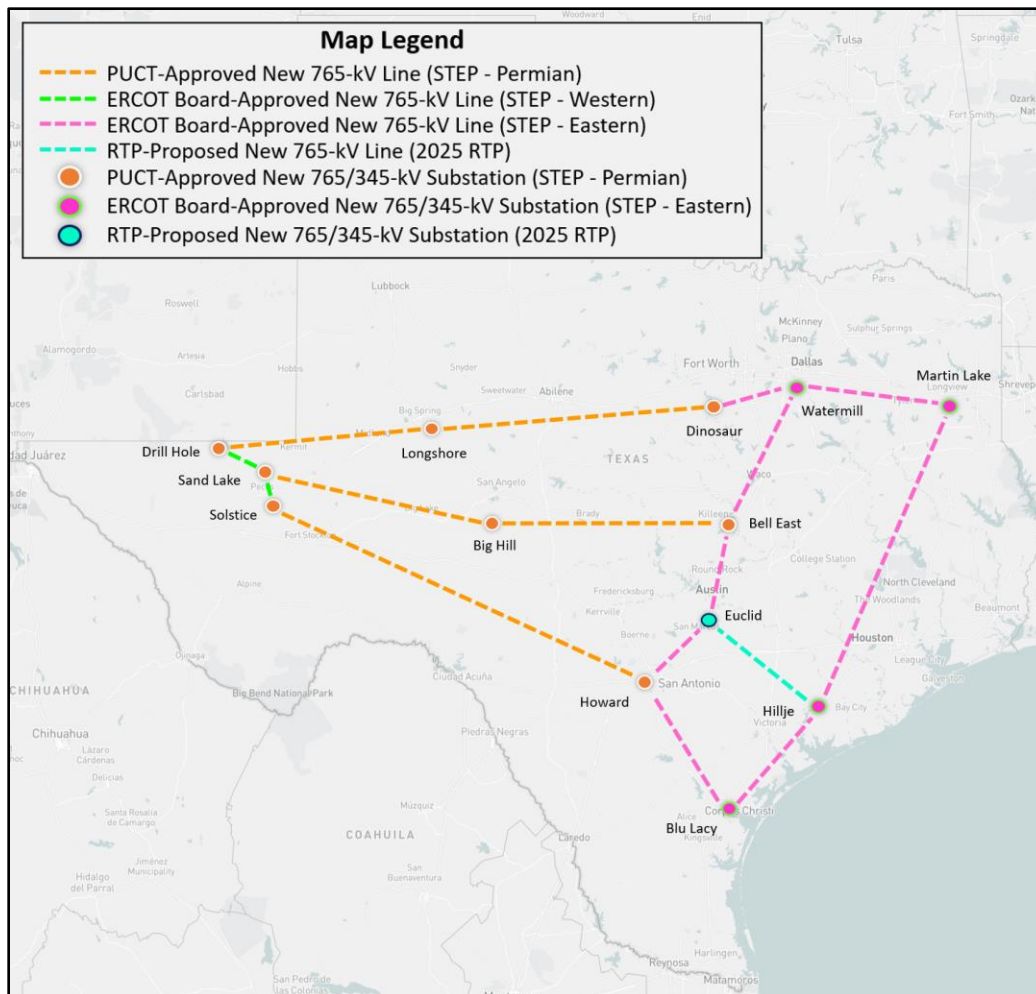


Figure 17: Additional 765-kV Equipment Identified in the 2025 RTP to Serve Growing Load in Central Texas

The expansion of the existing 345-kV Euclid station in Central Texas to include 765-kV helps stabilize bus voltages and provides a new source into the Central Texas area, while the approximately 130 mile Euclid to Hillje 765-kV line addresses various violations along the Coast-to-Central paths. Adding a new 765-kV Central Texas import pathway from the Hillje substation in the Coast Weather Zone to the Euclid station in Central Texas brings approximately 2,000 MW across the line into Central Texas under base case summer peak conditions. The increased import capability is observed under both normal and post-contingency conditions across various system topology configurations, including both

with the planned 765-kV STEP Eastern Backbone project and exclusively on the new line's own merits without the planned backbone as demonstrated in the early study years.

In addition to dramatically increasing overall import into the Central Texas area, the additional 765-kV pathway will provide the much-needed operational flexibility for the system to handle any required outage of those existing pathways without compromising system reliability.

The need for, and benefit from, this new 765-kV import path is seen as early as the 2027 study case. However, the total system load in the 2027 study case, which includes TSP-submitted large loads adjusted through the ETPALF process, is approximately 120,000 MW, approximately 35,000 MW higher than the current summer peak demand record of 85,508 MW. The 2028 study case, with TSP-submitted large loads, is approximately 140,000 MW, or 55,000 MW higher than the current summer peak demand record.

Lastly, it is understood that the in-service date for a new 765-kV line would not be feasible by 2027. However, based on those TSP-submitted load levels, an additional 765-kV import path was the best solution to address the system needs identified at those particular load levels, independent of the in-service timeline. Furthermore, similar alternative projects, including upgrading substantial miles of 345-kV lines or adding substantial new 345-kV Right-of-Way would experience similar in-service date challenges. Therefore, the planning approach used in the 2025 RTP was to present the preferred solution option. Should these system load levels materialize before the preferred transmission solution is completed, the TSPs and ERCOT will work collaboratively to determine an appropriate CMP for the operational situation at that time. Should those load levels materialize before the new import path is in place, approximately 3,000 MW of future load would need to have its energization delayed based on 2027 study case load levels, and 6,000 MW of future load would need to have its energization delayed based on 2028 study case load levels. Additional details can be found in Appendix G.

Detailed descriptions of all Central Texas area projects are in Appendix E. Expected future generation and load will likely impact these needs, and therefore ERCOT and TSPs will continue to evaluate the need and additional project options.

3.1.4 Other Findings

In addition to the reliability analysis summarized in previous sections, a multiple element outage analysis was conducted for contingencies where non-consequential load loss is allowed under NERC Reliability Standard TPL-001-5.1, Table 1. This analysis consisted of:

- Corrective action analysis, which identified mitigation measures (such as transformer tap setting changes, switching actions, generator re-dispatch, and load shed) to resolve any overloads and over/under-voltage issues resulting from such contingencies; and
- Cascading analysis, which identified any contingencies that could result in potential cascade events.

Some planning events and extreme events were screened for detailed analysis, and further investigation performed by ERCOT indicated that none of those events resulted in cascading

conditions. ERCOT also studied the loss of multiple generating stations due to the disruption of gas pipelines. The results of the multiple element outage analysis are documented in Appendix H. This appendix includes the list of critical contingencies identified as a result of this analysis and CAPs or recommendations necessary to mitigate the impact of these contingencies.

ERCOT also performed an analysis of known outages of generation and transmission facilities planned in the near-term planning horizon per Requirement 2.1.4 of NERC Reliability Standard TPL-001-5.1. ERCOT issued Market Notices¹⁸ to collect known outages information from both TOs and GOs. TOs were required to provide technical rationales for their selections of known outages for each study case included in the 2025 RTP. ERCOT developed the technical rationale for the selection of GO-submitted outages to be included in the corresponding study cases. The known outages from the TOs and the GO outages selected based on the ERCOT-developed technical rationale were then incorporated into the base cases to study their impact on system performance under P0 and P1 contingencies. The study results found one additional violation was caused by the known outages. The CAP developed to address this violation was project *2016-N01 – Kiamichi Energy Facility (140077) 345-kV Reactors Addition*, which was a CAP developed in the 2024 RTP under different system conditions. This CAP has been developed in many previous RTPs since 2016. Additionally, there is currently an active mitigation plan in ERCOT Operations to address the conditions observed that resulted in the CAP during the known outages analysis. When the RE performs the maintenance captured in this known outage analysis, the mitigation plan may need to be activated.

In addition to the above analyses, per ERCOT Planning Guide Section 3.1.1.2(3), the 2025 RTP analysis included a list of transmission facilities that were loaded above 95% of their applicable ratings under normal and contingency events (i.e., loss of single generating unit, transmission circuit, transformer, or common tower outage). This list is attached to the report as Appendix I.

3.2. Sensitivity (“Overnight”) Analysis

The unprecedented system load totals from the 2024 RTP that were again incorporated in the 2025 RTP necessitated going beyond the current planning practices to add generation to meet the demand. Portions of this generation did not meet Planning Guide 6.9(1) requirements. Stakeholders again raised the following two key questions in the 2025 RTP that were previously raised in the 2024 RTP:

1. How will system needs change if not all large loads materialize?
2. What are the impacts of the added generation that do not meet Planning Guide 6.9(1) requirements on system needs?

A third key question was raised in the 2025 RTP:

3. What are the impacts in the West and Far West Weather Zones of no solar and low wind system conditions?

¹⁸ https://www.ercot.com/services/comm/mkt_notices/W-B063025-01
https://www.ercot.com/services/comm/mkt_notices/W-A063025-01

The sensitivity analysis was designed to address the first and second key questions in the 2025 RTP, as it was in the 2024 RTP. In the sensitivity analysis, all large loads substantiated by TSP officer letters were reduced to zero and all generators not currently meeting Planning Guide 6.9(1) requirements were taken offline. The sensitivity analysis was performed on the 2027 and 2030 summer peak cases and the 2028 minimum load case. Despite reducing the load substantiated by TSP officer letters from over 23 GW to zero, the key reliability needs identified in the base case reliability analysis remained similar.

The 2028 minimum load case (the “Overnight Case”) was designed to answer the third key question. The Overnight Case represents a typical spring season overnight timeframe of approximately 2 a.m. to 4 a.m., when the system load is at its lowest annual point and solar and batteries are not dispatching at any level. A combination of dispatchable and wind generation is serving system load. Table 1 below summarizes key assumptions of the Overnight Base (BASE) case and the Overnight Sensitivity (SEN) case. The Overnight BASE case included Officer Letter large loads, generation that had not met the Planning Guide 6.9(1) requirements, and had wind dispatched at a higher level. Whereas the Overnight SEN case included only large loads with a signed interconnection agreement, generation that had met Planning Guide 6.9(1) requirements, and had wind dispatched to a much lower level.

Table 1: Summary of Overnight Sensitivity Case Assumptions

	Overnight BASE	Overnight SEN
Study Year	2028	2028
Load Scenario	Minimum Load	Minimum Load
Includes only Generation Meeting Planning Guide 6.9	YES	YES
System Load (Less Self-Served, MW)	83,518	71,736
Large Loads with TSP Officer Letters (System, MW)	11,782	0
Large Loads with Signed Interconnection Agreements (System, MW)	26,373	26,373
Far West Load (MW)	12,458	11,708
West Load (MW)	4,109	3,126
Far West Dispatchable Generation (MW)	2,446	2,446
West Dispatchable Generation (MW)	680	680
Far West Wind Generation (MW)	3,558	1,730
West Wind Generation (MW)	4,299	2,097
Far West Solar Generation (MW)	0	0
West Solar Generation (MW)	0	0
WFW Net Import Needs (MW)	5,584	7,880 (+2,296)
Solar Dispatch Level (%)	0	0
Battery Dispatch Level (%)	0	0
Wind Dispatch Level (Southern Coastal, %)	36	18 (-18)
Wind Dispatch Level (Southern Non-Coastal, %)	48	24 (-24)
Wind (Panhandle, %)	54	30 (-24)
Wind (Other, %)	48	24 (-24)

The total line project mile needs in Table 2 show that even though the Overnight SEN case had a greater net import need, the Overnight BASE case needed more new line addition miles and upgrade miles compared to the SEN case. This highlights that total load in the region plays a crucial role in the reliability of the area in addition to import need.

Table 2: Summary of Overnight Sensitivity Case WFW Line Project Needs

	Overnight BASE	Overnight SEN
345-kV WFW Line Project Needs (Additions, miles)	173	92
345-kV WFW Line Project Needs (Upgrades, miles)	240	192

Figures 18 and 19 show the geographical similarities of the 345-kV line project needs, both additions and upgrades, in the Overnight BASE and SEN cases, respectively. However, there is a difference in total magnitude of need.

These results reinforce what the Permian Basin Reliability Plan concluded, which is the Permian Basin area has a two-fold need: import capability and local network reinforcement.

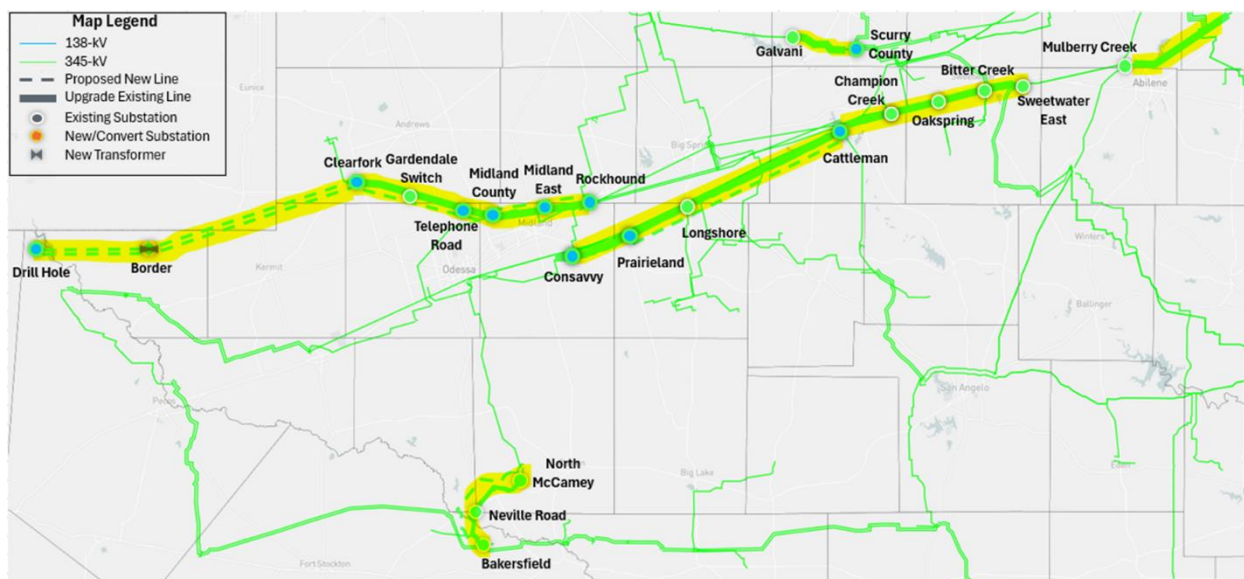


Figure 18: 345-kV Line Project Needs Seen in the 2028 Overnight Base Case in the West and Far West Weather Zones

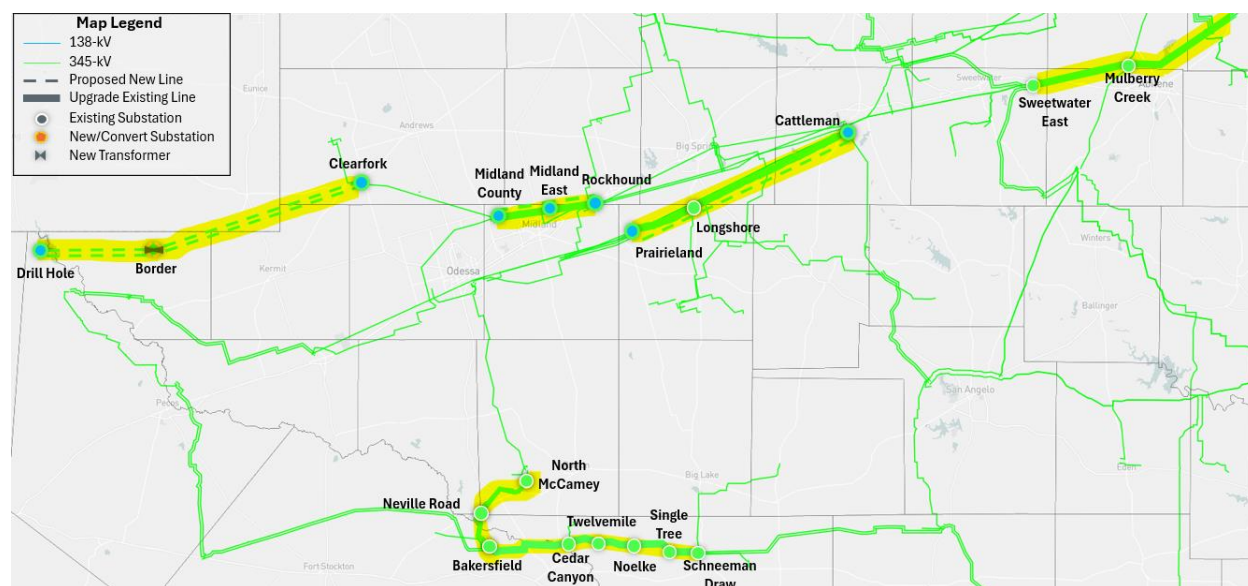


Figure 19: 345-kV Line Project Needs Seen in the 2028 Overnight Sensitivity Case in the West and Far West Weather Zones

The detailed assumptions and study results are available in Appendices B, E, and J, respectively.

3.3. Peak Net Load (“Sunset”) Analysis and Minimum Deliverability

To meet the Minimum Deliverability criteria as required by Planning Guide 4.1.1.7, the 2025 RTP developed a 2027 study case to represent “sunset” conditions typically found during the 7-8 p.m. time period.

The battery dispatch of 56.9% was based on the capacity factor observed on October 10, 2025, hour ending (HE) 19 when ERCOT set a battery discharge record of 8,714 MW.

The study condition was further stressed by mimicking a high Southern wind export condition. The wind and solar capacity factors used in the Sunset BASE case were from August 11, 2023, HE20, when real-time system conditions saw a 345-kV South Texas export line reach 123.2% under a potential contingency loss. These stressful system conditions were similar to September 6, 2023, when ERCOT entered Energy Emergency Alert conditions. The purpose of this analysis was not to focus solely on the South Texas export challenge because the future planning project to resolve that issue has already been ERCOT RPG-approved and will be energized at the point in time at which this study evaluates. Instead, this analysis was meant to capture what are still the most challenging operating conditions when solar is very low, wind in the West and Far West is low, and wind in the Panhandle is very low. These conditions resulted in generation from the Southern and Coast Weather Zones needing to travel across the entire system, including to the West and Far West Weather Zones, stressing many paths along the way.

The generation in this case includes only future generation that meets Planning Guide requirements 6.9(1). The load in this study case is approximately 106 GW and contains only large loads with signed interconnection agreements. This case does not include Officer Letter loads. The system load level is

95.8% of the summer peak load level found in the RTP Sensitivity BASE case discussed in Section 3.2. The Sensitivity BASE case was the starting point for the Sunset analysis since the Sensitivity BASE case did not include Officer Letter loads and included only Planning Guide 6.9(1) generation.

Table 3 below summarizes the load and generation details of the Sunset Base (BASE) case and the Sunset Minimum Deliverability (MIN DELIV) case. The Sunset BASE was developed with the load and generation assumptions shown in Table 3, and then an N-1 contingency analysis was performed, and projects were developed to resolve violations found. Then, analysis involving a prior outage of a generator or 345/138-kV transformer followed by N-1 contingency conditions was performed, and projects were developed to resolve issues found.

The Sunset BASE case with its N-1 projects and prior outage projects then had its generation updated to reflect the Minimum Deliverability criteria to become the Sunset MIN DELIV case. The analysis was then repeated on the adjusted Sunset MIN DELIV case. N-1 contingency analysis was performed, and projects were developed to resolve the violations that resulted. This was followed by analysis involving a prior outage of a generator or 345/138-kV transformer followed by N-1 contingency conditions. Projects were developed to address newly identified issues.

The purpose of first making the Sunset BASE case secure under the above conditions was to ensure the violations seen after making the Minimum Deliverability adjustments were a direct result of those generation adjustments. Those generation adjustments included setting Generation Resources utilizing combined cycle, steam turbine, combustion turbine, hydro, or reciprocating engine technology to 100% of their output along with any Energy Storage Resource (ESR) with a duration greater than or equal to two hours. For ESRs with a duration of less than two hours, a prorated deliverability was ensured. The output of these generator types was then fixed and redispatch of those generator types was not allowed. This contrasts with the Sunset BASE case, where dispatchable units were allowed to reach 100% of their possible output, but with redispatch enabled, some units were curtailed to prevent violations. This resulted in an effective dispatch of approximately 94% for dispatchable units in the Sunset BASE case.

Table 3: Summary of Sunset Base Case and Sunset Minimum Deliverability Assumptions

	Sunset BASE	Sunset MIN DELIV
Study Year	2027	2027
Load Scenario	Peak Net Load	Peak Net Load
Includes only Generation Meeting Planning Guide 6.9	YES	YES
System Load (Less Self-Served, MW)	106,487	106,487
Percent of Summer Peak Load Level	95.8	95.8
Large Loads with TSP Officer Letters (System, MW)	0	0
Large Loads with Signed Interconnection Agreements (System, MW)	19,271	19,271
Dispatchable Generation Level (%)	100 (94) ¹⁹	100
Solar Dispatch Level (%)	13.3	5.7
Battery Dispatch Level (%)	56.9	83.3
Wind Dispatch Level (Southern Coastal, %)	63.3	31.1
Wind Dispatch Level (Southern Non-Coastal, %)	70.9	34.7
Wind (Panhandle, %)	12.0	6.1
Wind (Other, %)	33.6	14.4

¹⁹ Dispatchable generators in the Sunset BASE case could be redispatched to resolve violations, which resulted in an effective dispatch of 94%. Dispatchable generators were not allowed to be redispatched in the Sunset MIN DELIV case and were maintained at 100% output.

Analysis of the Sunset BASE case revealed significant 345-kV line project needs in all study regions. Figures 20-23 show those needs. Green pathways highlighted in yellow represent 345-kV projects identified in the Sunset BASE analysis, while purple pathways highlighted in yellow indicate 345-kV needs identified by the Minimum Deliverability analysis. Figure 24 illustrates the flow from the Coast, Southern, and South Central Weather Zones to the West and Far West Weather Zones in both the Sunset BASE and Sunset MIN DELIV cases.

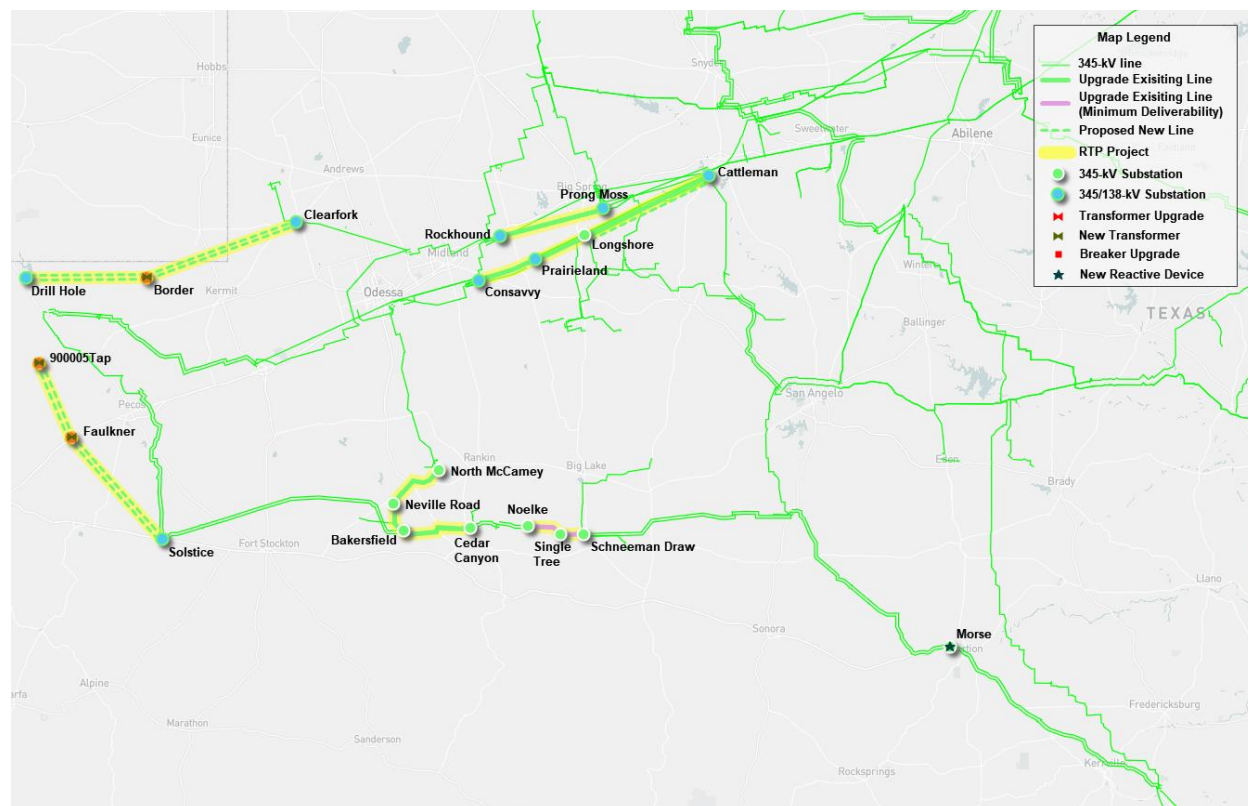


Figure 20: 345-kV Line Project Needs Seen in the 2027 Sunset Case (Yellow) and 2027 Sunset Minimum Deliverability Case (Purple) in the West and Far West Weather Zones

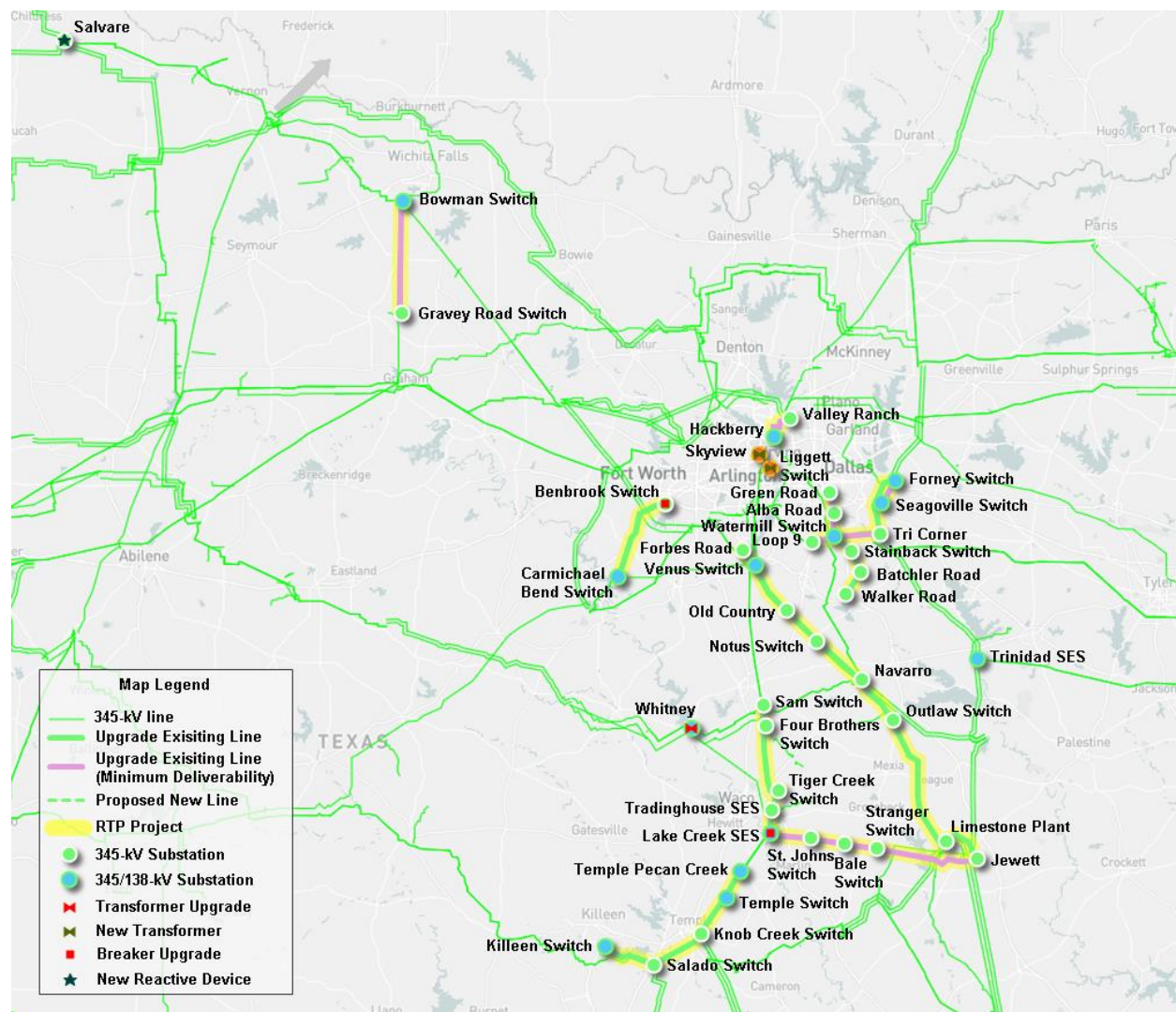


Figure 21: 345-kV Line Project Needs Seen in the 2027 Sunset Case (Yellow) and 2027 Sunset Minimum Deliverability Case (Purple) in the North and North Central Weather Zones

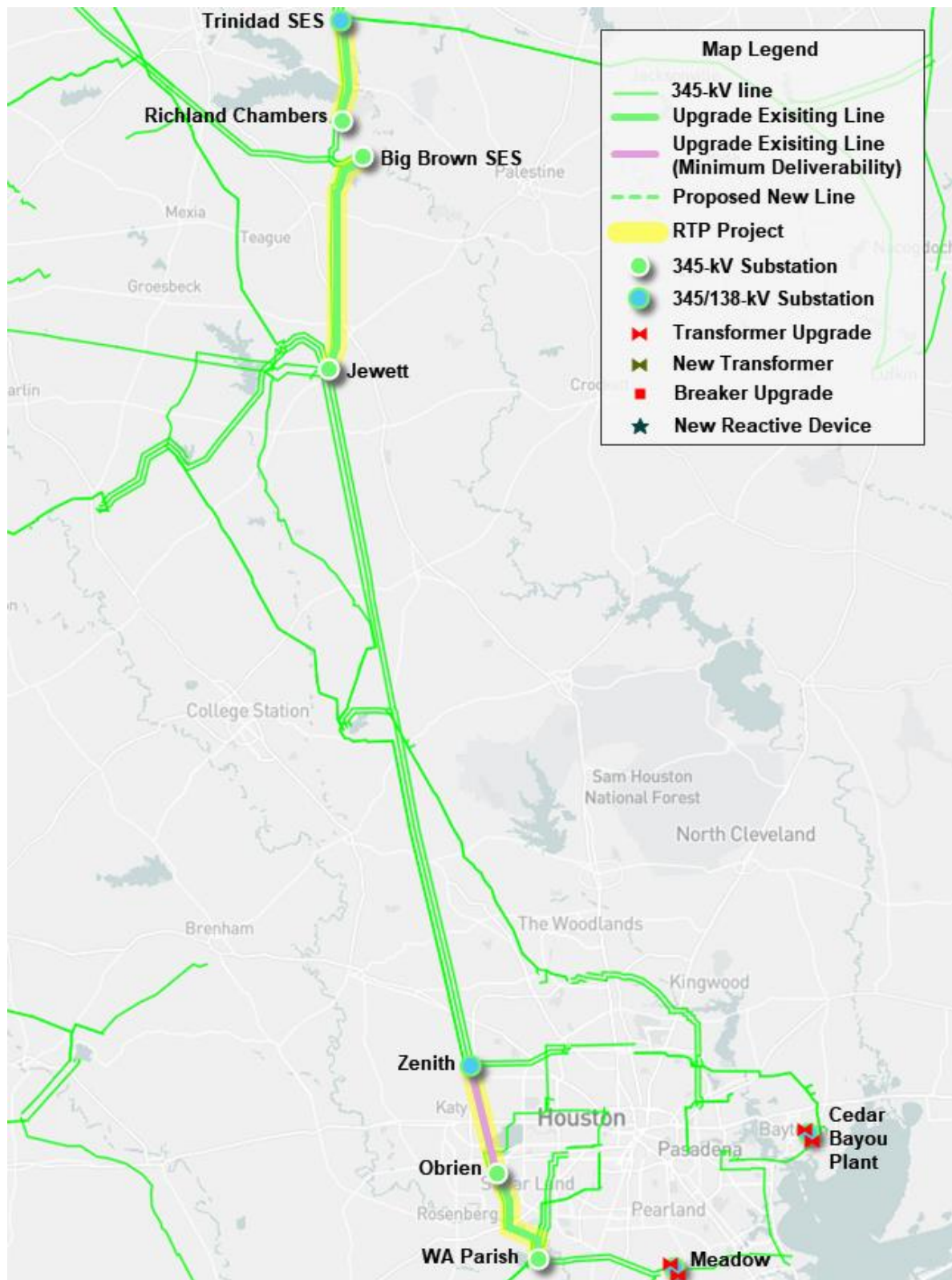


Figure 22: 345-kV Line Project Needs Seen in the 2027 Sunset Case (Yellow) and 2027 Sunset Minimum Deliverability Case (Purple) in the East and Coast Weather Zones

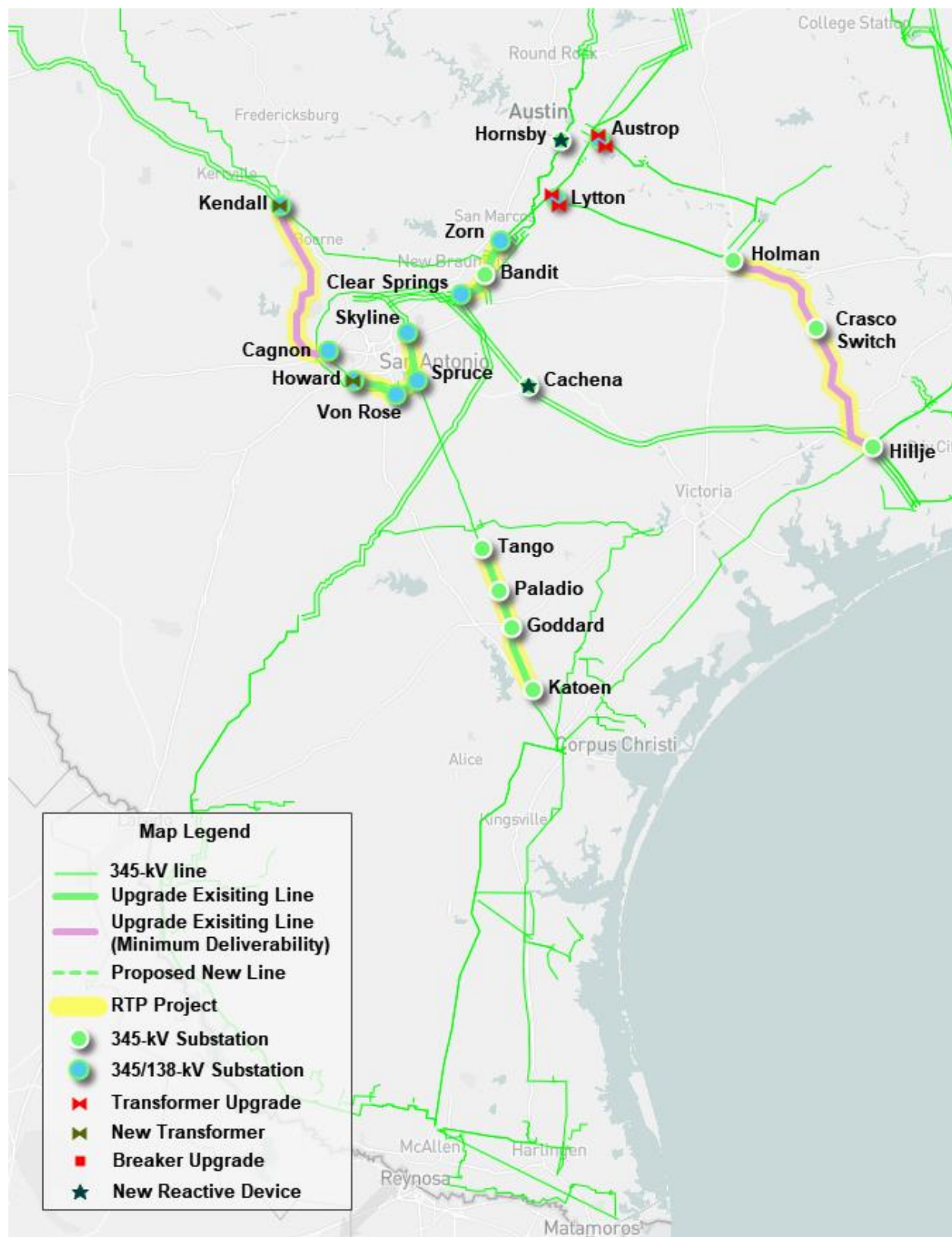


Figure 23: 345-kV Line Project Needs Seen in the 2027 Sunset Case (Yellow) and 2027 Sunset Minimum Deliverability Case (Purple) in the Southern and South Central Weather Zones

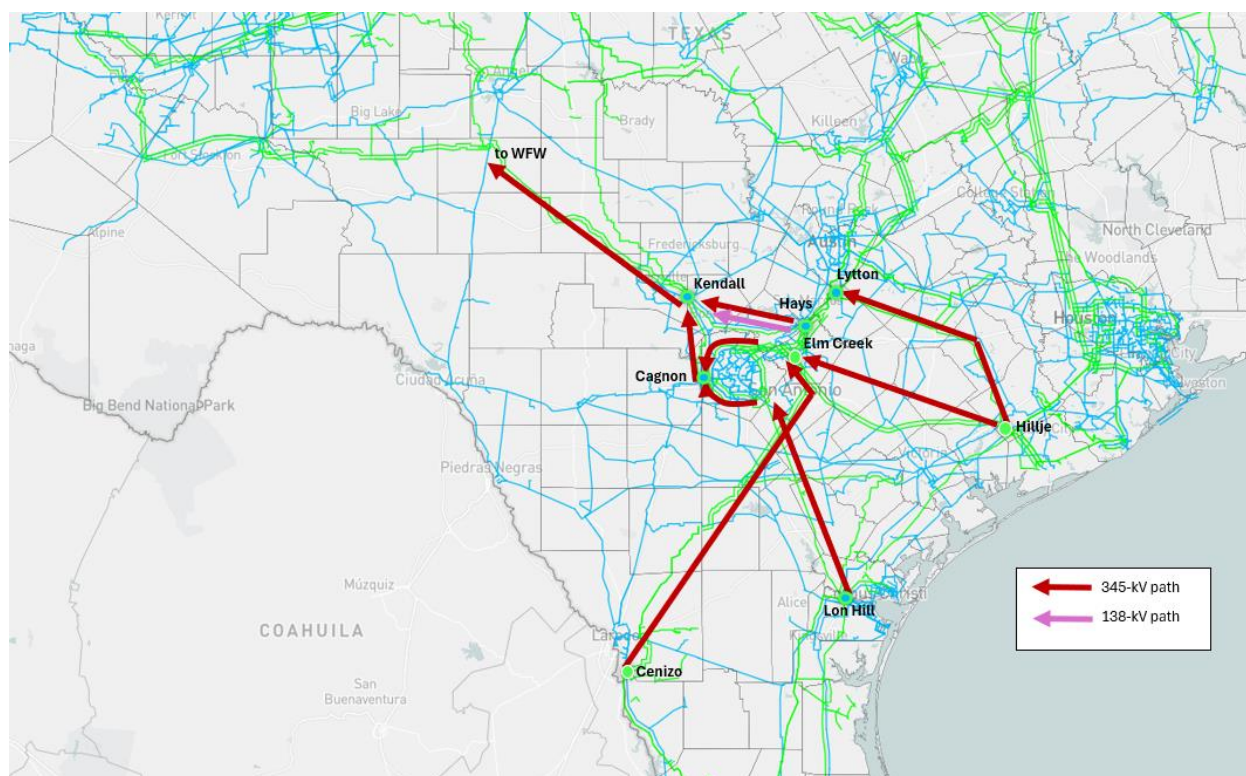


Figure 24: Flow Patterns Seen in the 2027 Sunset Case and 2027 Sunset Minimum Deliverability Case

3.4. Short Circuit Analysis

As indicated in Section 2.4, ERCOT conducted a short circuit analysis for RE-owned facilities for 2028 summer peak conditions based on the system protection future year base case and shared the results with GOs. GOs reviewed the fault duty information to identify buses with over-dutied breakers and CAPs.

Table 4 provides a summary of the results of the short circuit analysis. The study cases and details of the results can be found in Appendix K.

Table 4: Summary of Short Circuit Analysis

Magnitude of Fault Current	Number of buses (3-phase fault)	Number of buses (single-line-to-ground fault)
Below 40 kA	654	676
40 kA ~ 60 kA	97	72
More than 60 kA	3	6

3.5. Long Lead Time Equipment Analysis

In response to ERCOT's request, TOs provided a list of long lead time equipment based on their spare equipment strategies. All TO-provided BES long lead time equipment outages were studied to determine the impact of unavailability of such equipment for an extended period of time. This analysis was conducted for 2027 and 2030 summer peak conditions, along with 2028 off-peak conditions. Overall, 36 unique 345/138-kV transformers, three unique 345/115-kV transformers, 21 unique 345-kV reactive devices, 29 unique reactive devices at other voltage levels, four 345-kV synchronous condensers and two of their transformers, three unique 345-kV SVCs, four unique 138-kV SVCs, two unique 138-kV STATCOMs, and two unique GIS circuit breakers were identified as long lead time equipment. NERC category P0, P1, and P2 planning events were studied. The results were shared with the respective TPs. The list of long lead time equipment and study results are provided in Appendix L.

4. Economic Analysis

The 2025 RTP economic analysis was performed using production cost simulation runs for years 2027 and 2030. In the analysis, both the production cost savings test and the congestion cost savings test were utilized. Through this assessment, ERCOT identified transmission congestion and tested various transmission improvements to address the congestion in a cost-effective manner as defined by ERCOT's economic planning criteria.

A transmission system upgrade is economically viable if it meets the 1) production cost savings test requirement, or 2) congestion cost savings test requirement.

Twelve economic transmission improvement projects were evaluated in the 2025 RTP. Break-even capital cost was provided for the projects that showed positive economic benefits.

Details of the economic analysis and project evaluation results can be found in Appendix P.

4.1. Study Assumptions and Methodology

The 2025 RTP economic cases used the 2024 RTP final reliability cases as the start cases, specifically the 2027 summer peak case and the 2030 summer peak case with the 765-kV - STEP incorporated.

4.1.1 Load and Generation Assumptions

Two sets of cases with different load and generation assumptions were developed for each study year to demonstrate the system impact under different forecasted load levels. The first set of cases included both the large loads with a signed Interconnection Agreement and the large loads substantiated by TSP Officer Letters with the PUCT-approved GCE applied²⁰. This set of cases is referred to as base cases in this report, and the resulting coincident peak for study year 2030 is approximately 123.6 GW. The other set of cases included only the large loads with a signed Interconnection Agreement. This set of cases is referred to as sensitivity cases in this report, and the resulting coincident peak for study year 2030 is approximately 109.6 GW.

Similar to the reliability analysis, ERCOT took steps outside of the normal planning process to incorporate more Generation Resources into the planning models used in the 2025 RTP economic analysis based on the rules proposed by PGRR127²¹. PGRR127 modifies the criteria for adding Generation Resources to the planning models and is currently going through the stakeholder process. The generators were added in the sequence outlined below until there was sufficient generation to meet the demand for all hours:

- Large generators that have a signed Standard Generation Interconnection Agreement submitted by the TSP via the online Resource Integration and Ongoing Operations system;
- Large generators that have completed the FIS; and
- Large generators that have started the FIS.

²⁰ https://www.ercot.com/files/docs/2025/06/17/2025_RTP_Load_Update_June_2025_RPG.pdf

²¹ <https://www.ercot.com/mktrules/issues/PGRR127>

The large load and generation assumptions for both the base and sensitivity cases are listed in Table 5 and Table 6, respectively.

Table 5: Large Load Assumptions for Base and Sensitivity Cases

	Base Cases	Sensitivity Cases
Large Loads	1. Large load profiles from 2024 RTP were used 2. Contracted loads were included at 100% and Officer Letter loads (OLLs) were adjusted as follows. 2.1. OLLs were reduced to 55.4% 2.2. Data Center OLLs are further reduced to 49.8% of the initial OLL reduction	1. Large load profiles from 2024 RTP were used 2. Contracted loads were included at 100% and all OLLs were removed

Table 6: Generation Assumptions for Base and Sensitivity Cases

	Base Cases	Sensitivity Cases
Added Generation Capacity	Added from the May 2025 Generator Interconnection Status (GIS) report 1. Planning Guide Section 6.9(1) 2. IA signed 3. FIS Completed 4. FIS Started (partial)	Added from the May 2025 Generator Interconnection Status (GIS) report 1. Planning Guide Section 6.9(1) 2. IA signed 3. FIS Completed (partial)

4.1.2 Interface Limit Assumptions

With the three new 765-kV Permian Basin import paths, the West Texas Export (WTX) interface limit is expected to increase from 12.7 GW to 16.2 GW²². The WTX interface limits for the two study years can be found in Table 7.

Table 7: WTX Interface Limits

Interface	2027 Case	2030 Case
WTX Interface Limit (MW)	12,700	16,200

The ERCOT Board of Directors endorsed the Lower Rio Grande Valley (LRGV) System Enhancement Project in 2021. The PUCT also ordered the construction of a new second circuit on the double-circuit capable 345-kV transmission line that runs from San Miguel to Palmito and new transmission facilities to close the loop from Palmito to North Edinburg. These projects plan to be in operation before the summer of 2027. The North Edinburg to Lobo and Nelson Sharpe to Rio Hondo interfaces were retired in the 2025 RTP economic analysis for both 2027 and 2030 due to those improvements.

²² <https://www.ercot.com/files/docs/2025/01/27/2024-regional-transmission-plan-rtp-345-kv-plan-and-texas-765-kv-strategic-transmission-expans.pdf>

The South Texas Export and Import interfaces were first modeled in the 2024 RTP economic analysis. Due to the following transmission upgrades²³, the South Texas Export and Import interfaces were retired in the 2025 RTP economic analysis.

- San Antonio South Reliability Project (endorsed, projected in-service date: 2027)
- LRGV System Enhancement Project (endorsed, projected in-service date: 2027)
- San Antonio South Reliability II Project (endorsed, projected in-service date: 2030)
- Tango-Goddard-Katoen-Lon Hill 345-kV line second circuit (part of the South to Central Texas project from the 2023 RTP).

4.2. System Summary for 2027 and 2030 Study Years

Table 8 provides a system summary of the 2025 RTP economic analysis for years 2027 and 2030. The input assumptions and the input data for this study can be found in Appendices C and D respectively.

Table 8: System Summary of 2027 and 2030²⁴

Description	Unit	2027 Base	2030 Base	2027 Sensitivity	2030 Sensitivity
Coincident Peak Load	MW	110,123	123,559	101,896	109,593
Peak Net Load ²⁵	MW	88,549	104,613	81,700	89,630
Annual Served Demand	GWh	701,537	831,233	629,563	708,866
Annual Storage Charging	GWh	24,472	27,555	20,792	25,981
Annual Transmission Losses	GWh	16,028	14,861	15,396	14,361
Annual Generation	GWh	742,036	873,649	665,751	749,209
Load-Weighted Average Locational Marginal Price (LMP)	\$/MWh	24.4	30.6	22.2	28.6

4.3. Wind/Solar Curtailment

While there may be hours when all ERCOT demand could theoretically be served only by wind and solar resources, thermal and stability constraints on the transmission system and resource operational limitations may cause the production cost simulation software to curtail available wind and solar power output. Figures 25 to 28 summarize monthly energy production and curtailment for wind and solar generation, respectively. In the base cases, 3.5% of wind energy and 6.7% of solar energy were curtailed in 2027, while 2.1% of wind and 3.9% of solar energy were curtailed in 2030. For the

²³ https://www.ercot.com/files/docs/2024/07/25/05-gtc_update_stx-export-and-import-aug1-ros-3.pdf

²⁴ All results are based on the 2013 historical weather conditions

²⁵ Hourly Net Load = Hourly Load Forecast + Hourly Battery Charging Load – Hourly Wind Output – Hourly Solar Output. The battery charging load at the peak net load hour is 0 MW.

sensitivity case, 5.5% of wind and 10.0% of solar energy were curtailed in 2027, and 3.3% of wind and 6.7% of solar energy were curtailed in 2030.

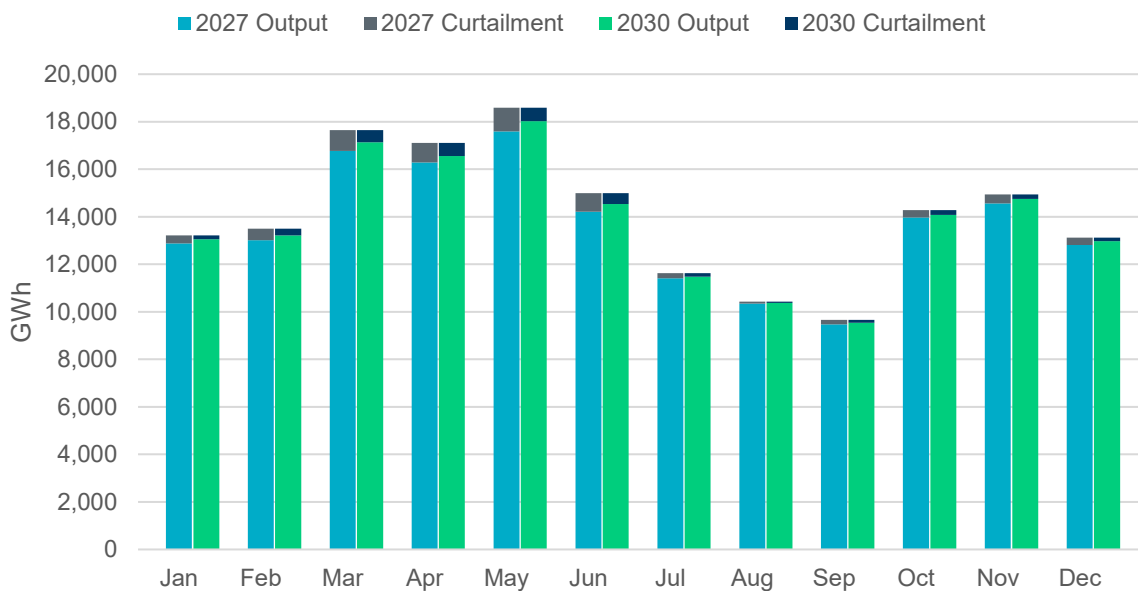


Figure 25: Wind Production and Curtailment (Base Case)

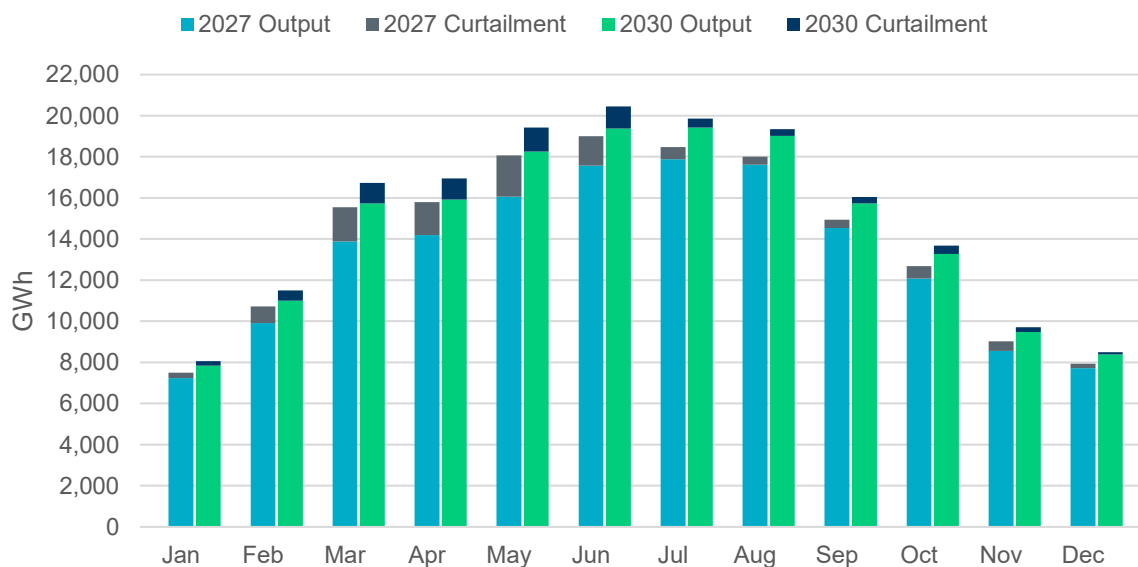


Figure 26: Solar Production and Curtailment (Base Case)

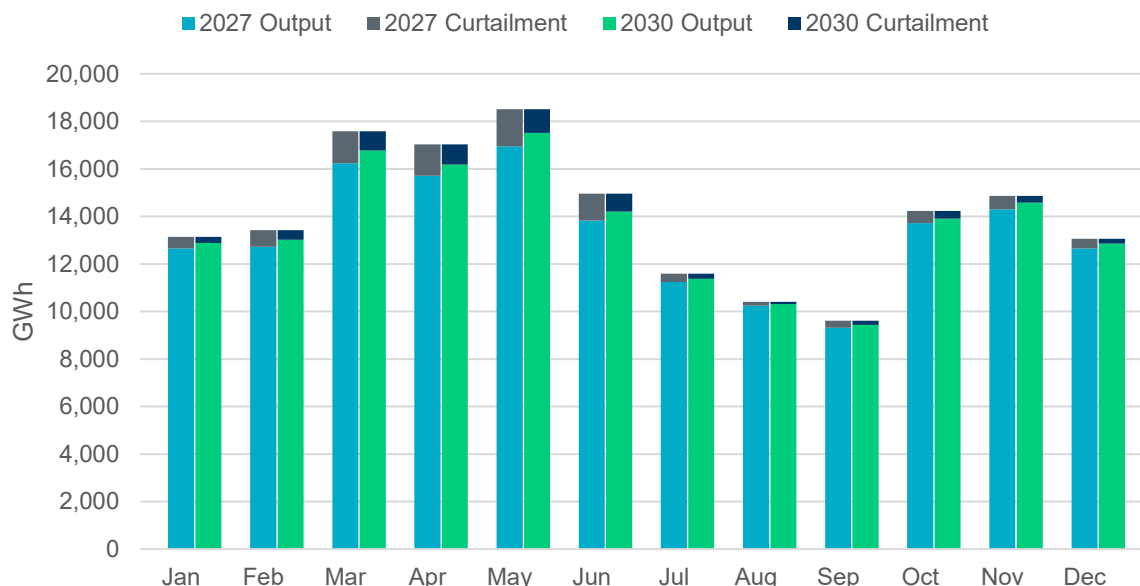


Figure 27: Wind Production and Curtailment (Sensitivity Case)

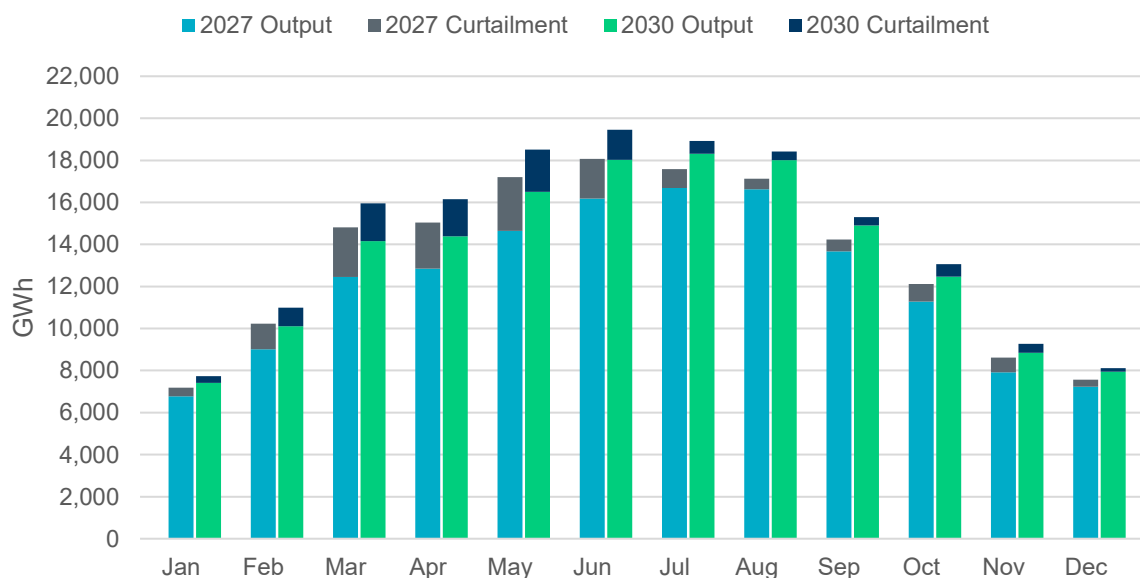


Figure 28: Solar Production and Curtailment (Sensitivity Case)

4.4. Top Constraints for 2027 and 2030 Study Years

The total congestion rent for the 2027 base and sensitivity cases was \$1.17 billion and \$1.11 billion, respectively, while the total congestion rent for the 2030 base and sensitivity cases was \$729 million and \$851 million, respectively.

Table 9 shows the projected top 10 constraints ranked by the congestion rent on the ERCOT System based on the economic analysis conducted for the 2027 and 2030 base cases. Figure 29 shows locations of the constraints listed in Table 9. Table 10 shows the projected top 10 constraints for the 2027 and 2030 sensitivity cases. Figure 30 shows locations of the constraints listed in Table 10.

The annual congestion, as well as monthly congestion details for this study can be found in Appendix D.

Table 9: Top Constraints for 2027 and 2030 Base Cases

Map	Constraint	Congestion Rent (2027 Base)	Congestion Rent (2030 Base)
1	West Texas Export interface	\$130M	\$147M
2	Houdini to WA Parish 345-kV line	\$155M	\$5M
3	Bell County East Switch to Voss Lake 345-kV line	\$112M	-
4	Panhandle interface	\$19M	\$72M
5	Holly POI to Wett Long Draw 345-kV line	\$18M	\$69M
6	North to Houston interface	\$26M	\$60M
7	Fowlerton to Tilden 138-kV line	\$33M	\$33M
8	Murray to Paint Creek 138-kV line	\$56M	-
9	Farmland to Holly POI 345-kV line	\$47M	\$9M
10	MacKenzie Substation to Northeast Substation 115-kV line	\$48M	\$6M

Table 10: Top Constraints for 2027 and 2030 Sensitivity Cases

Map	Constraint	Congestion Rent (2027 Sensitivity)	Congestion Rent (2030 Sensitivity)
1	West Texas Export interface	\$245M	\$209M
2	North to Houston interface	\$48M	\$141M
3	Houdini to WA Parish 345-kV line	\$150M	\$22M
4	Bell County East Switch to Voss Lake 345-kV line	\$106M	-
5	Holly POI to Wett Long Draw 345-kV line	\$9M	\$48M
6	Panhandle interface	\$12M	\$45M
7	Farmland to Holly POI 345-kV line	\$35M	\$13M
8	Elmato POI to Houdini 345-kV line	\$33M	\$14M
9	Buntin Drive to Simpson Stuart 138-kV line	-	\$37M
10	Oasis to Savana POI 345-kV line	\$17M	\$20M

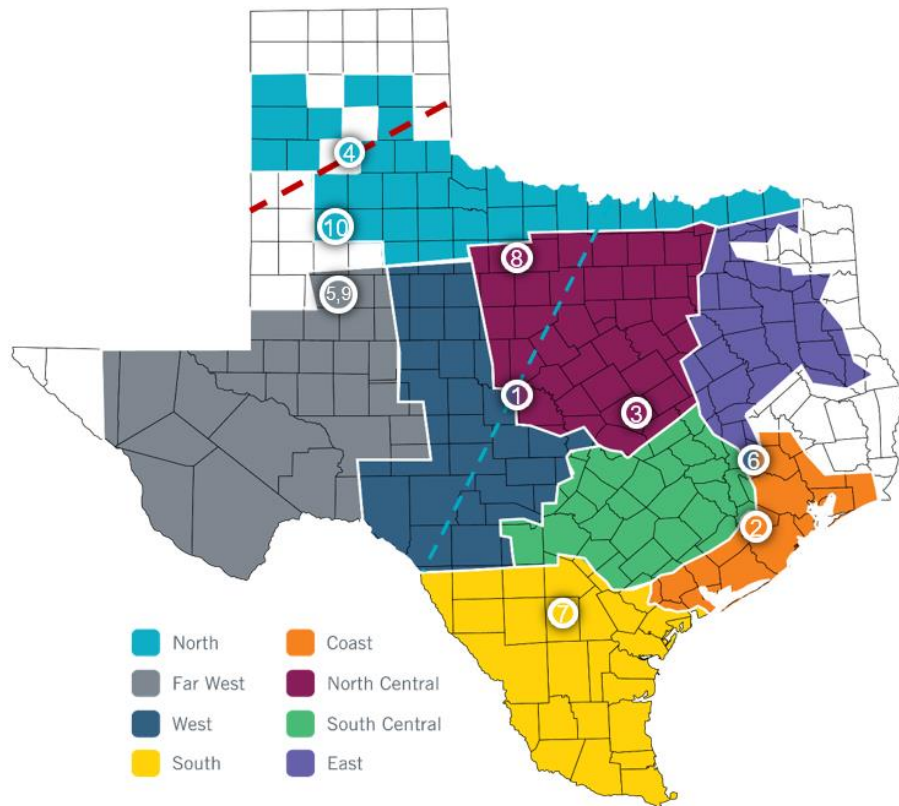


Figure 29: Locations of Top 10 Constraints in 2027 and 2030 Base Cases

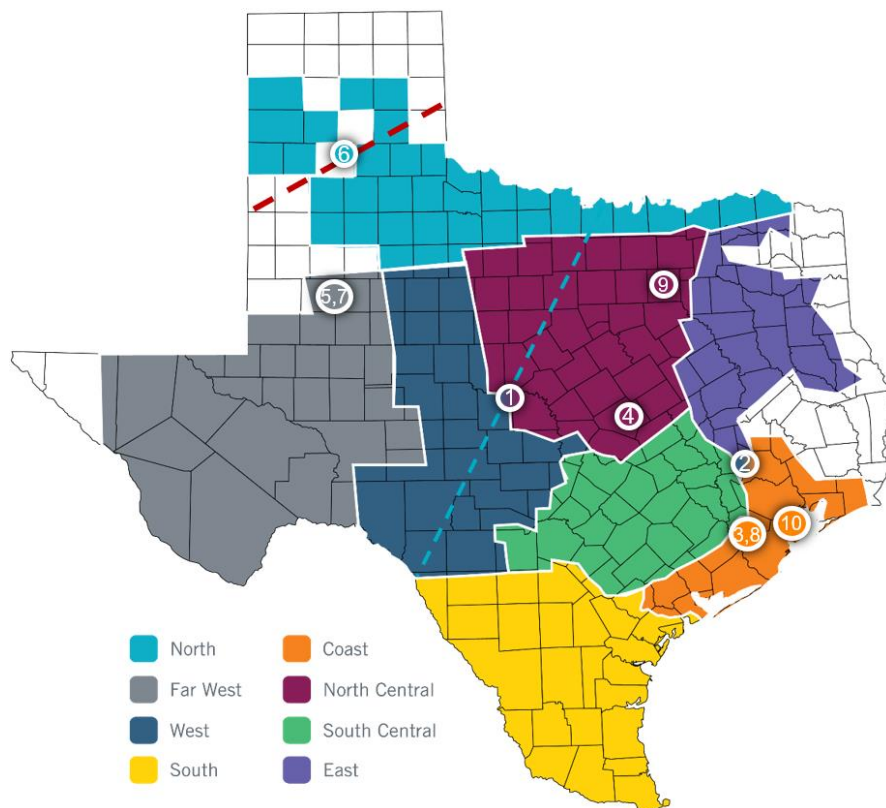


Figure 30: Locations of Top 10 Constraints in 2027 and 2030 Sensitivity Cases

4.4.1 Top Constraints for Sensitivity Cases

The 2025 RTP economic analysis used the 2027 and 2030 sensitivity cases to assess the cost-effectiveness of potential transmission improvements for top constraints. Brief descriptions of the top congestion seen in these cases are presented below, and more details can be found in Appendix P.

In the West and Far West Weather Zones, the rapid growth in renewable generation is expected to continue. Like the 2024 RTP economic analysis, the WTX interface was the most congested element observed for both the 2027 and 2030 study years. The interface was congested approximately 24.1% and 15.8% of hours in the 2027 and 2030 study years, respectively. The congestion rent on this interface was \$245 million in 2027. With the 765-kV STEP incorporated in the 2030 study case, the congestion rent for the WTX interface remained high at \$209 million. Opportunities for further transmission investment to increase the interface limit should be monitored in the future as renewable generation buildout continues in the West, Far West, and Panhandle regions.

The North to Houston interface²⁶ showed a congestion rent of \$48 million in 2027 and \$141 million in 2030. The interface was congested 11.8% of hours in 2027 and 25.6% of hours in 2030 due to rising energy demand in the greater Houston area.

The Panhandle area is also seeing continuous development of new renewable resources, resulting in \$12 million and \$45 million congestion rent in 2027 and 2030, respectively, on the Panhandle interface. Moreover, in the same area, the Wett Long Draw to Holly POI to Farmland 345-kV line was also among the top constraints.

The Houdini to WA Parish 345-kV line was projected to incur high congestion rent in 2027 (\$150 million). The congestion rent dropped to \$22 million in 2030 with the 765-kV STEP incorporated. The percentage of congestion time dropped from 24.3% of hours in 2027 to 6.7% of hours in 2030.

Based on the review of the initial congestion pattern and historical congestion, ERCOT selected transmission projects to conduct both the production cost savings test and the congestion cost savings test. A summary of the evaluated projects is provided in Section 4.5.

4.5. Projects Selected for Evaluation

The congestion patterns observed for the 2027 and 2030 study years in sensitivity cases served as the basis for identifying potential economic projects for further evaluations. In addition to the projected congestion outlined above, ERCOT also considered historical congestion observed in those areas and took experiences with these constraints in past economic studies into consideration to select several additional projects for further evaluation.

Table 11 shows the list of all transmission projects evaluated in this economic analysis. Figure 31 shows the location of each project. Detailed descriptions of these projects are included in Appendix P.

²⁶ The North to Houston interface limit for the 2025 RTP economic analysis was based on the voltage stability limit found using VSAT for peak summer conditions, then scaled to hourly profiles from historical data.

Table 11: Projects Selected for Economic Evaluation

Index	Description
Project 1	Elmato POI to Houdini to WA Parish 345-kV line upgrade
Project 2	Friendswood to Webster and Friendswood to Mustang Bayou 138-kV line upgrades
Project 3	Fisher Road Switch to Short Creek Solar POI 345-kV line and Fisher Road Switch to Wichita Falls 138-kV line upgrade
Project 4	Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line upgrade
Project 5	Bell County East to Voss Lake 345-kV double-circuit line upgrade
Project 6	Four Brothers Switch to Tradinghouse SES to Lake Creek 345-kV line upgrade
Project 7	Mackenzie Substation to Northeast Substation to Dunbar Substation 115-kV line upgrade
Project 8	Carver to Poblano 69-kV to 138-kV line conversion
Project 9	Sulphur Springs Switch to Liberty to Edgewood 138-kV line upgrade
Project 10	Fowlerton area improvement
Project 11	White River to Long Draw 345-kV double-circuit line addition
Project 12	North to Houston import improvement

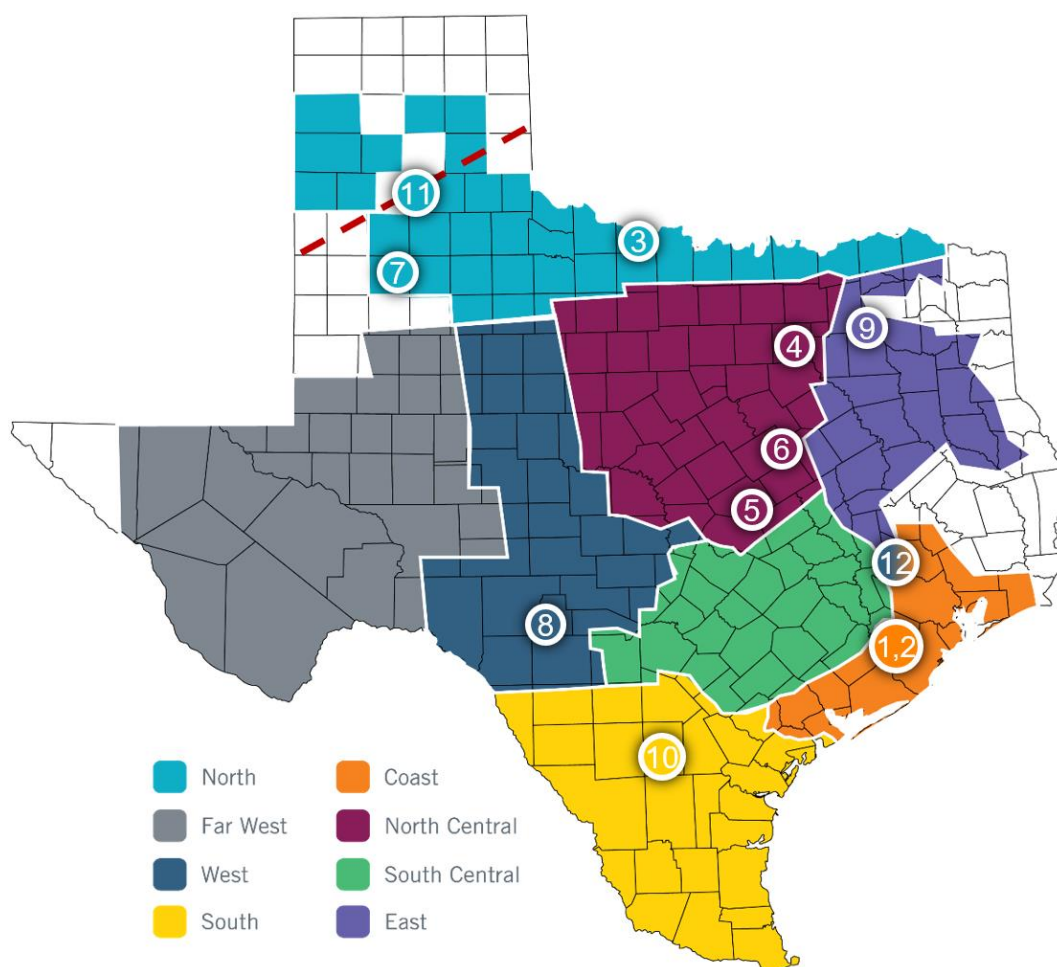


Figure 31: Proposed Project Locations

4.6. Project Evaluation Results

Using the 2025 RTP sensitivity cases for both 2027 and 2030 study years, ERCOT performed an economic analysis for 12 transmission enhancements chosen based on a review of the initial congestion patterns and historical data.

Both the congestion cost savings test, measured by the system-wide consumer energy cost reduction, and the production cost savings test were performed in the project evaluation. The total production cost is the sum of the fixed operation and maintenance (O&M), startup, variable O&M, fuel, and emission cost of generators. The total consumer energy cost equals the sum of load payments for energy use. To pass the production cost savings test, the levelized ERCOT-wide annual production cost savings attributable to the proposed project should be equal to or greater than the first-year annual revenue requirement (13.0%) of the proposed project. To satisfy the congestion cost savings test requirement, the levelized ERCOT-wide annual consumer energy cost reduction attributable to the proposed project should be equal to or greater than the average of the annual revenue requirement for the first three years (12.7%) of the proposed project. These revenue requirements are reviewed

annually and may vary from year to year. ERCOT may recommend a transmission upgrade in the ERCOT Region that passes either the production cost savings test or the congestion cost savings test.

The key findings are as follows:

- Based on the generic cost estimates, the following projects can potentially meet the economic planning criteria. However, the ultimate determination of the economic viability of the proposed transmission projects should be based on detailed cost estimates provided by the TSPs.
 - Friendswood to Webster and Friendswood to Mustang Bayou 138-kV line upgrades (project 2)
 - Fisher Road Switch to Short Creek Solar POI 345-kV line and Fisher Road Switch to Wichita Falls 138-kV line upgrade (project 3)
 - Buntin Drive to Simpson Stuart to Witt Road to Green Road 138-kV line upgrade (project 4)
- Accelerating the in-service date of the Tier 4 Carver to Poblano 69-kV to 138-kV line conversion project (project 8) to 2027 provided significant economic benefits.

Details of the economic analysis and the project evaluation results can be found in Appendix P.

Finally, as required by ERCOT Protocols Section 3.10.8.4(3), ERCOT identified additional transmission elements that have a high probability of providing significant added economic efficiency to the ERCOT market using dynamic ratings. Dynamic ratings for the identified elements (listed in Appendix M) have been requested from the associated TOs.

5. Appendices

Index	Description	Document	Access
A	RTP Scope and Process Document	Appendix_A_2025_RTP_Scope_and_Process_Final.pdf <file included in the public version>	Public
B	Input assumptions for the 2025 RTP reliability analysis	Appendix_B_2025_RTP_Reliability_Input_Assumptions.xlsx <file included in the public version>	Public
C	Input assumptions for the 2025 RTP economic analysis	Appendix_C_2025_RTP_Economic_Input_Assumptions.xlsx <file included in the public version>	Public
D	Economic analysis start case inputs and annual/monthly constraints	Appendix_D_2025_RTP_Economics_Start_Case_Inputs_and_Constraints.zip <file available in MIS Secure Area>	MIS Secure
E	Reliability Driven Projects	Appendix_E_2025_RTP_Reliability_Projects_Public.xlsx <file included in the public version>	Public
F	Project locations	Appendix_F_2025_RTP_Project_Locations.pdf <file included in the public version>	Public
G	Base Analysis Constraint Management Plans	Appendix_G_2025_RTP_Constraint_Management_Plans.xlsx <file available in MIS Secure Area>	MIS Secure
H	Multiple element contingency analysis	Appendix_H_2025_RTP_MultipleElementContingencyStudyReport.docx <file is ERCOT-Confidential>	N/A
I	Facilities loaded over 95%	Appendix_I_2025_RTP_95%_Exceedance_PG31123.xlsx <file available in MIS Secure Area>	MIS Secure
J	Sensitivity Analysis Results	Appendix_J_2025_RTP_Sensitivity_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
K	Short circuit Analysis	Appendix_K_2025_RTP_ShortCircuitStudyCases_DetailedResults.docx <file available in MIS Secure Area>	MIS Secure
L	Long lead time equipment analysis	Appendix_L_2025_RTP_LongLeadTimeEquipment.docx <file is ERCOT-Confidential>	N/A
M	Transmission elements proposed to be dynamically rated	Appendix_M_2025_RTP_Dynamic_Rating_NP3_10_8_4.xlsx <file available in MIS Secure Area>	MIS Secure
N	Sunset Base Analysis	Appendix_N_2025_RTP_Sunset_Base_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
O	Sunset Minimum Deliverability Analysis	Appendix_O_2025_RTP_Sunset_Minimum_Deliverability_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
P	Economic Analysis Results	Appendix_P_2025_RTP_Economic_Analysis_Results.pdf <file included in the public version>	Public

Q	Sensitivity Analysis Constraint Management Plans	Appendix_Q_2025_RTP_Sensitivity_Constraint_Management_Plans.xlsx <file available in MIS Secure Area>	MIS Secure
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